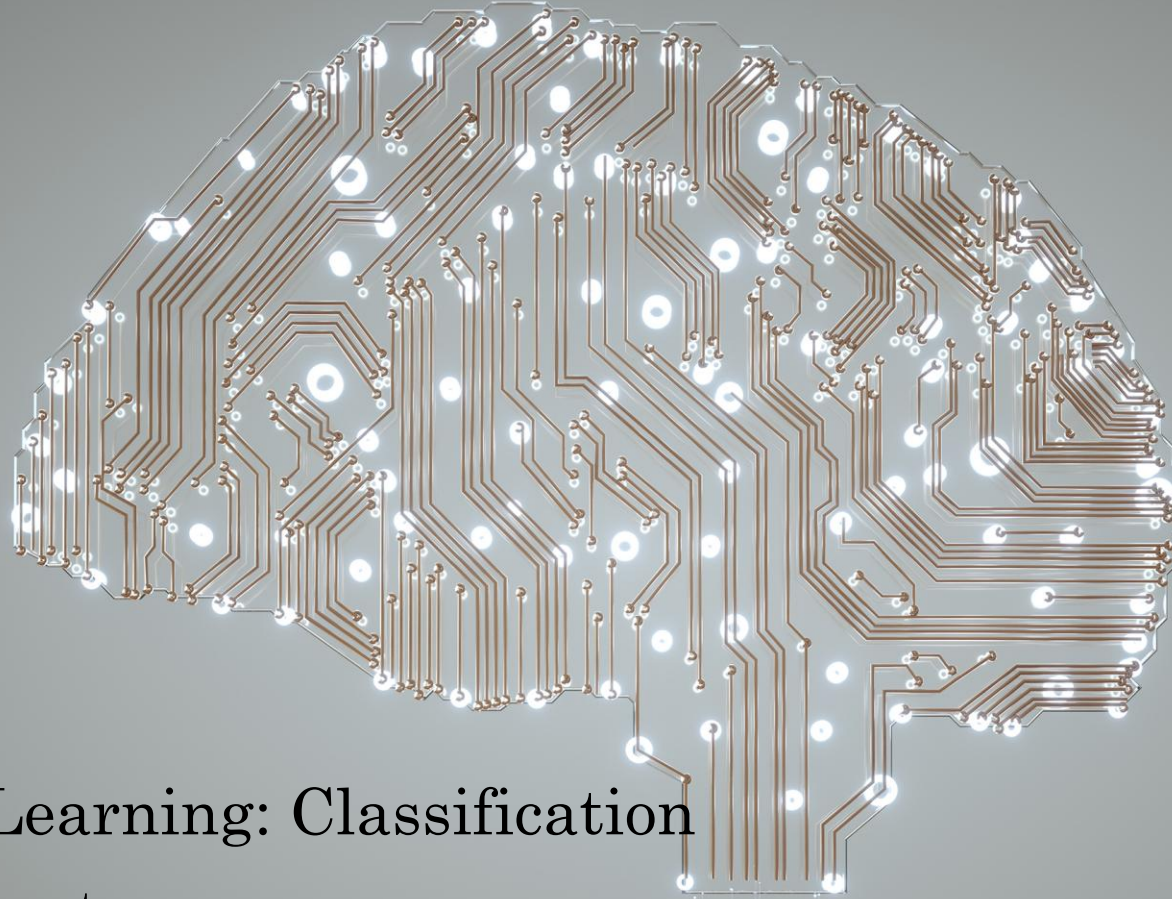


ML & NLP: PGR210



Supervised Learning: Classification

Dr. Arvind Keprate

Topics for Today

Supervised ML Workflow

What is Classification and How is it different from Regression?

K-Nearest Neighbour (KNN) Algorithm

Decision Tree Algorithm

Model Evaluation Metrics

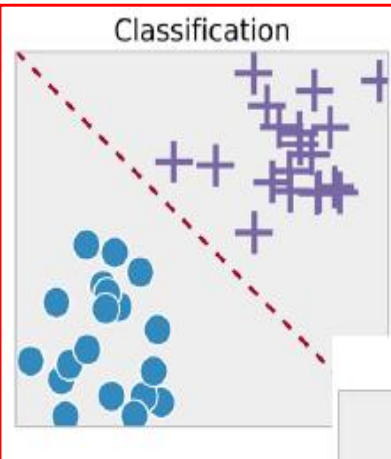
Problem Introduction

Types of ML?

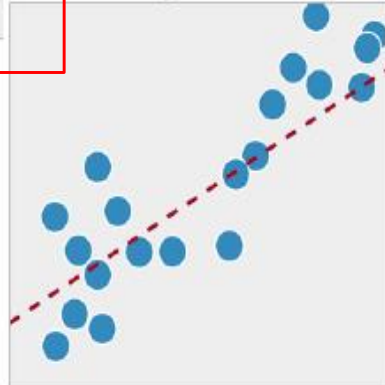
Machine Learning

1. Supervised Learning

Classification

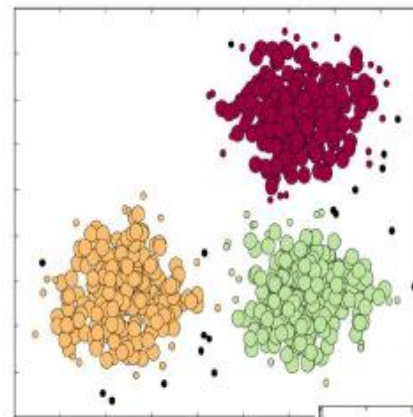


Regression

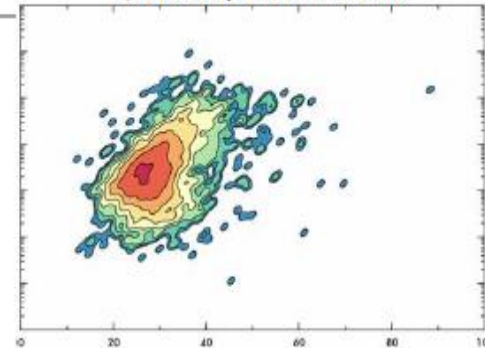


2. Unsupervised Learning

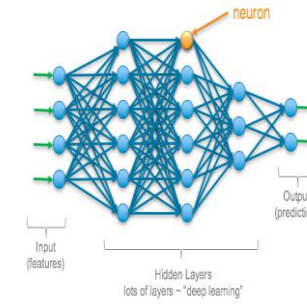
Clustering



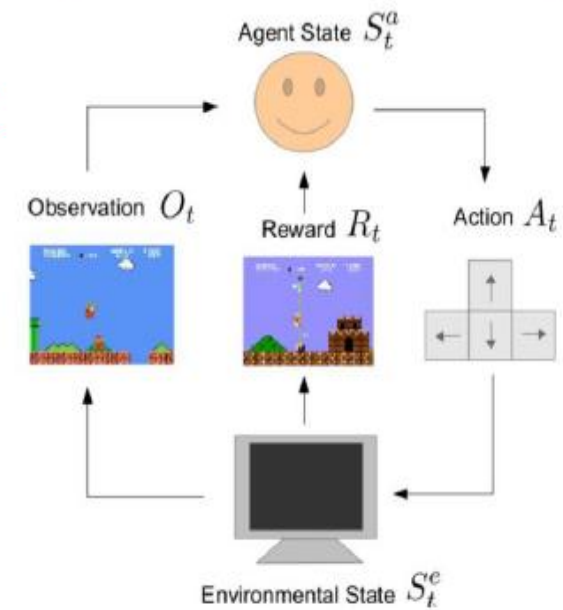
Density Estimation



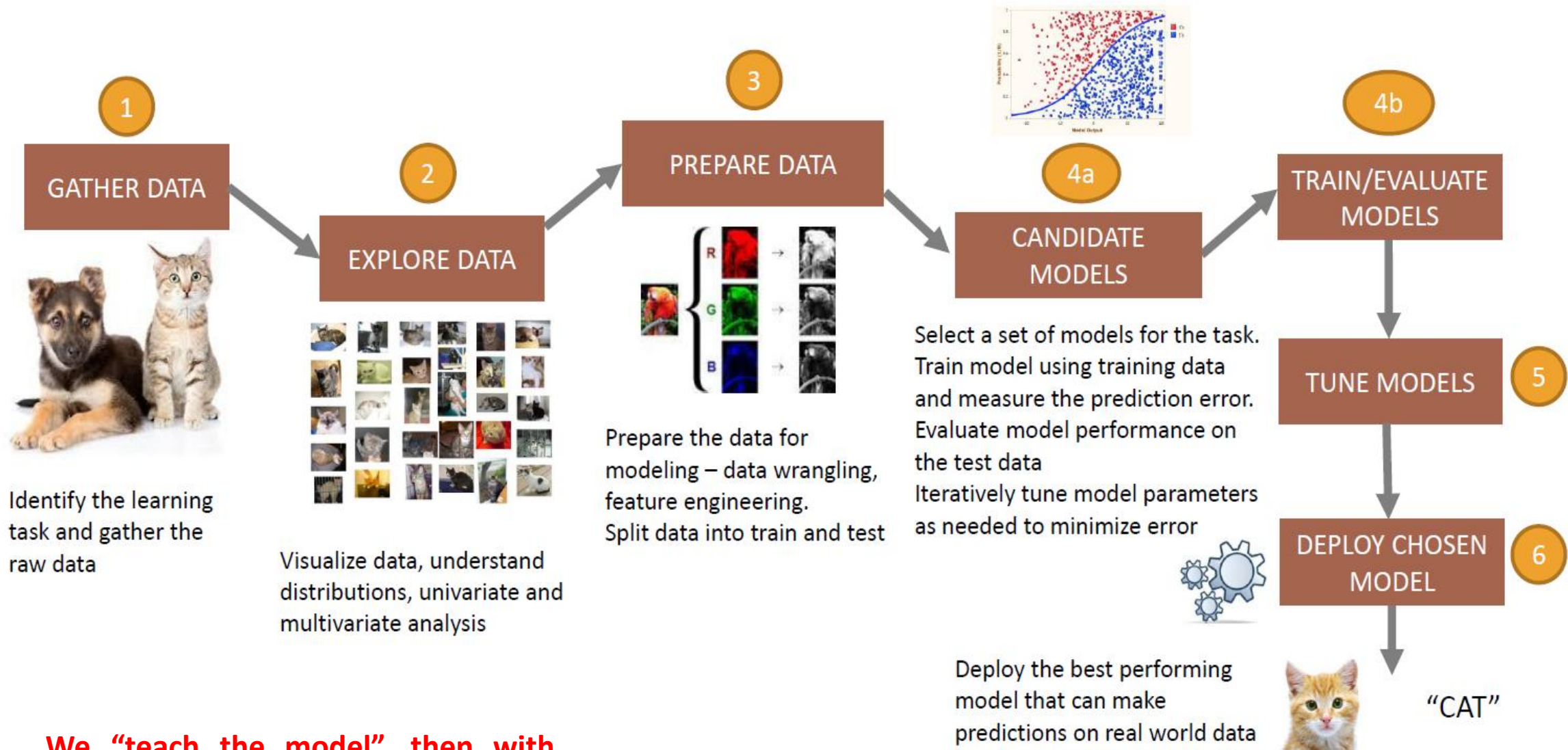
4. Deep Learning



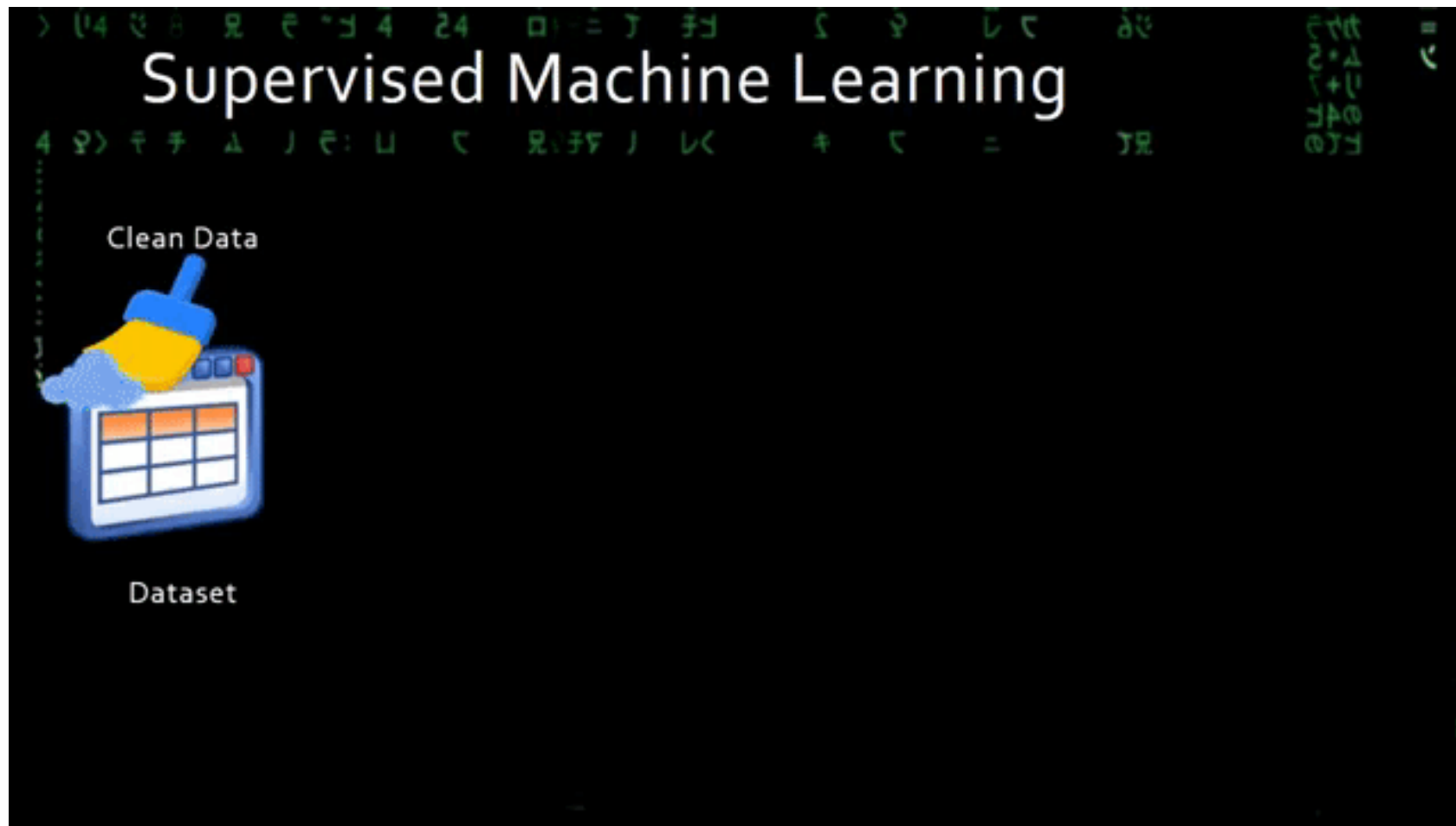
3. Reinforcement Learning



Supervised ML Workflow



We “teach the model”, then with that knowledge, it can predict unknown or future instances.



Supervised ML Workflow

Regression vs. Classification



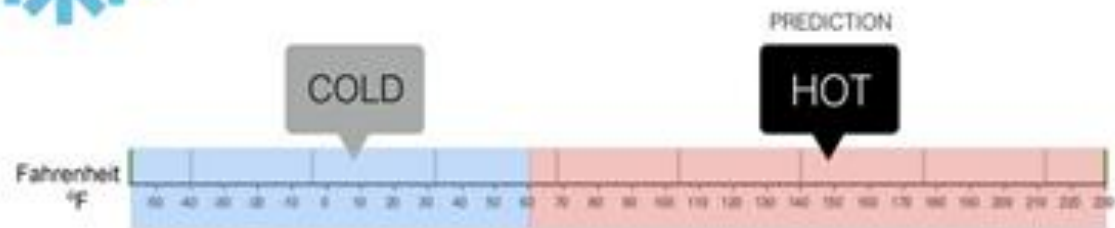
Regression

What is the temperature going to be tomorrow?



Classification

Will it be Cold or Hot tomorrow?



Continuous output space: regression

- **Pollution level prediction**
Predict tomorrow's air quality index in my neighborhood
 $\mathcal{Y} = [0, \infty)$ (< 100 : okay, > 200 : dangerous)
- **Insurance company calculations**
What is the expected life expectancy of this person?
 $\mathcal{Y} = [0, 120]$

Discrete output space: classification

Binary classification

E.g., Spam detection

$\mathcal{X} = \{\text{email messages}\}$

$\mathcal{Y} = \{\text{spam, not spam}\}$

Multiclass

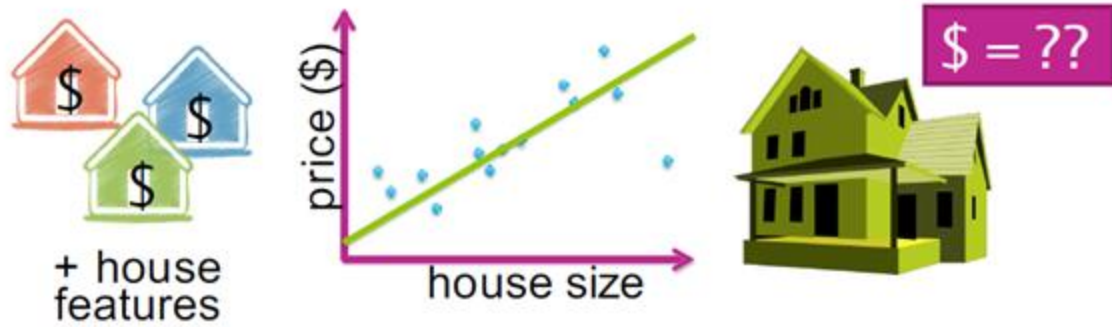
E.g., News article classification

$\mathcal{X} = \{\text{news articles}\}$

$\mathcal{Y} = \{\text{politics, business, sports, ...}\}$

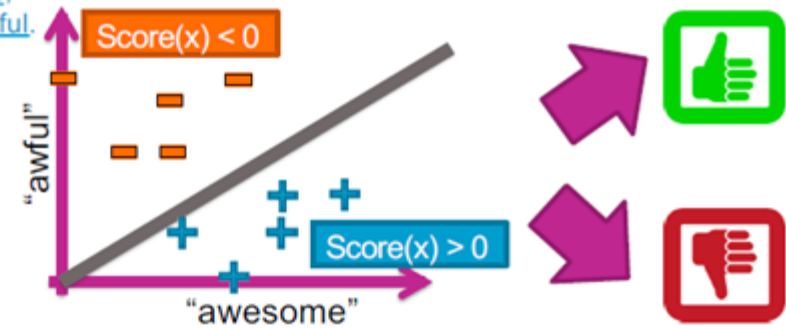
Trivia Time

Regression



Sushi was awesome,
the food was awesome,
but the service was awful.

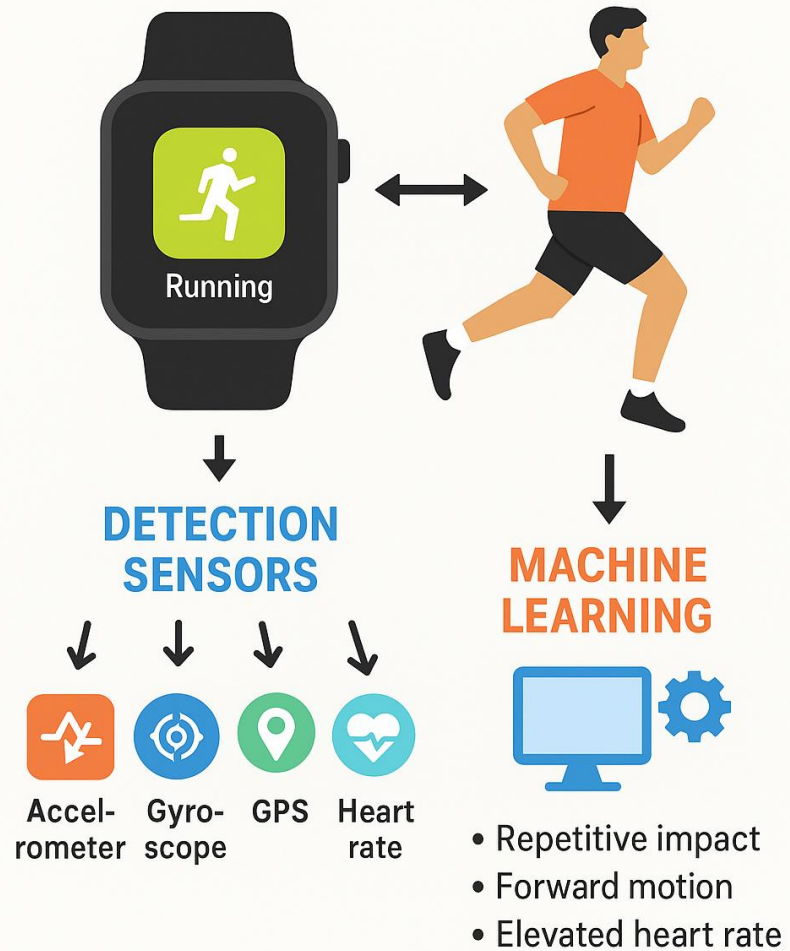
All reviews:



Classification

Trivia Time

HOW THE APPLE WATCH DETECTS RUNNING



Classification ★

Or

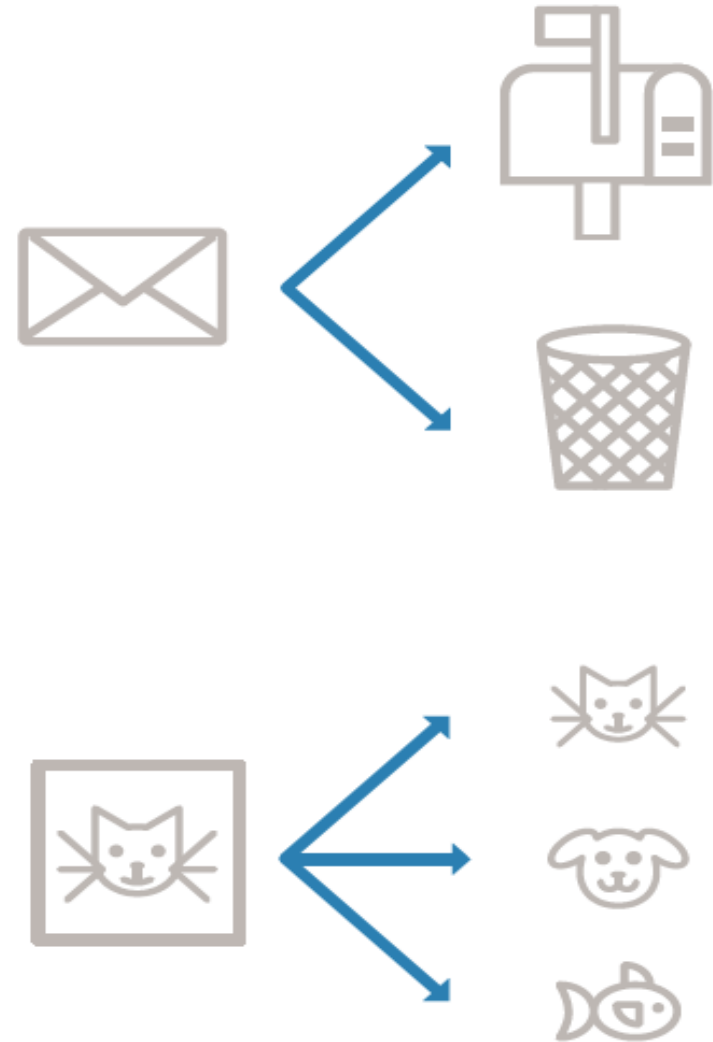
Clustering

Binary vs. Multiclass Classification

When you are working on a classification problem, begin by determining whether the problem is binary or multiclass. In a binary classification problem, a single training or test item (instance) can only be divided into two classes—for example, if you want to determine whether an email is genuine or spam. In a multiclass classification problem, it can be divided into more than two—for example, if you want to train a model to classify an image as a dog, cat, or other animal.

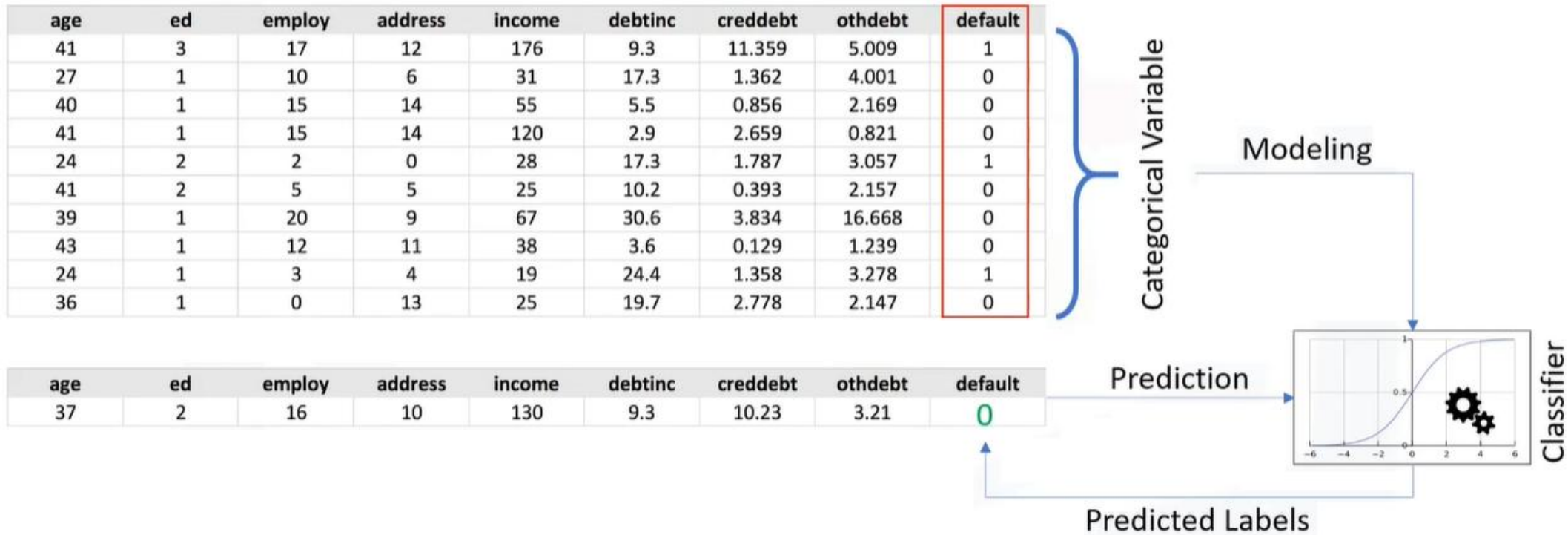
Bear in mind that a multiclass classification problem is generally more challenging because it requires a more complex model.

Certain algorithms (for example, logistic regression) are designed specifically for binary classification problems. During training, these algorithms tend to be more efficient than multiclass algorithms.



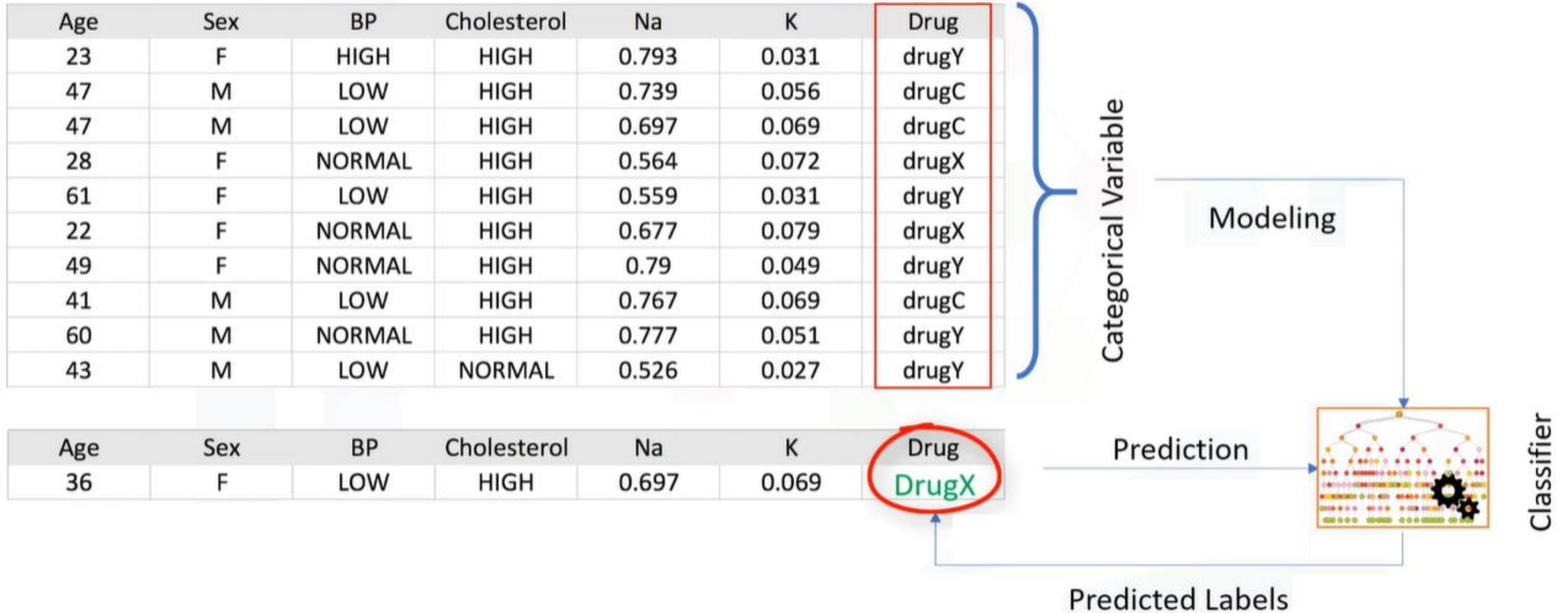
Binary Classification Example

- Classification determines the class label for an unlabeled test case.
- Loan Default Prediction Problem.



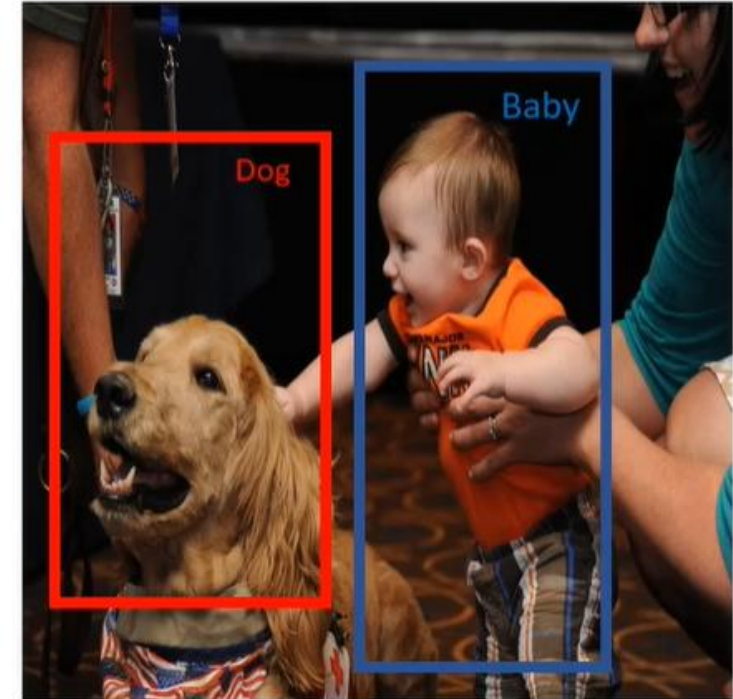
Example: Multi-Class Classification

- Drug Usage Prediction Problem.



Classification Applications & Algorithms

- Speech Recognition.
- Biometric Identification.
- Handwriting Recognition.



Some Algorithms:

- | | | |
|-----------------------------------|----------------------|-------------------------------|
| •Decision Trees (ID3, C4.5, C5.0) | •K-Nearest Neighbor | •Neural Network |
| •Naïve Bayes. | •Logistic Regression | •Support Vector Machine (SVM) |

Classification Algorithms

Algorithm	Prediction Speed	Training Speed	Memory Usage	Required Tuning	General Assessment
Logistic Regression (and Linear SVM)	Fast	Fast	Small	Minimal	Good for small problems with linear decision boundaries
Decision Trees	Fast	Fast	Small	Some	Good generalist, but prone to overfitting
Nearest Neighbor	Moderate	Minimal	Medium	Minimal	Lower accuracy, but easy to use and interpret
Naïve Bayes	Fast	Fast	Medium	Some	Widely used for text, including spam filtering
Ensembles	Moderate	Slow	Varies	Some	High accuracy and good performance for small- to medium-sized datasets
Neural Network	Moderate	Slow	Medium to Large	Lots	Popular for classification, compression, recognition, and forecasting

k Nearest Neighbor (kNN)

How it Works

kNN categorizes objects based on the classes of their nearest neighbors in the dataset. kNN predictions assume that objects near each other are similar. Distance metrics, such as Euclidean, city block, cosine, and Chebychev, are used to find the nearest neighbor.

Best Used...

- When you need a simple algorithm to establish benchmark learning rules
- When memory usage of the trained model is a lesser concern
- When prediction speed of the trained model is a lesser concern



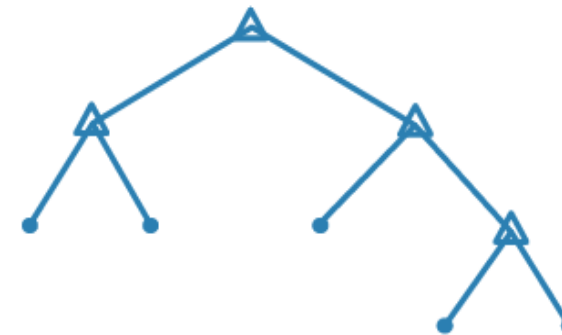
Decision Tree

How it Works

A decision tree lets you predict responses to data by following the decisions in the tree from the root (beginning) down to a leaf node. A tree consists of branching conditions where the value of a predictor is compared to a trained weight. The number of branches and the values of weights are determined in the training process. Additional modification, or pruning, may be used to simplify the model.

Best Used...

- When you need an algorithm that is easy to interpret and fast to fit
- To minimize memory usage
- When high predictive accuracy is not a requirement



k-Nearest Neighbour (KNN)

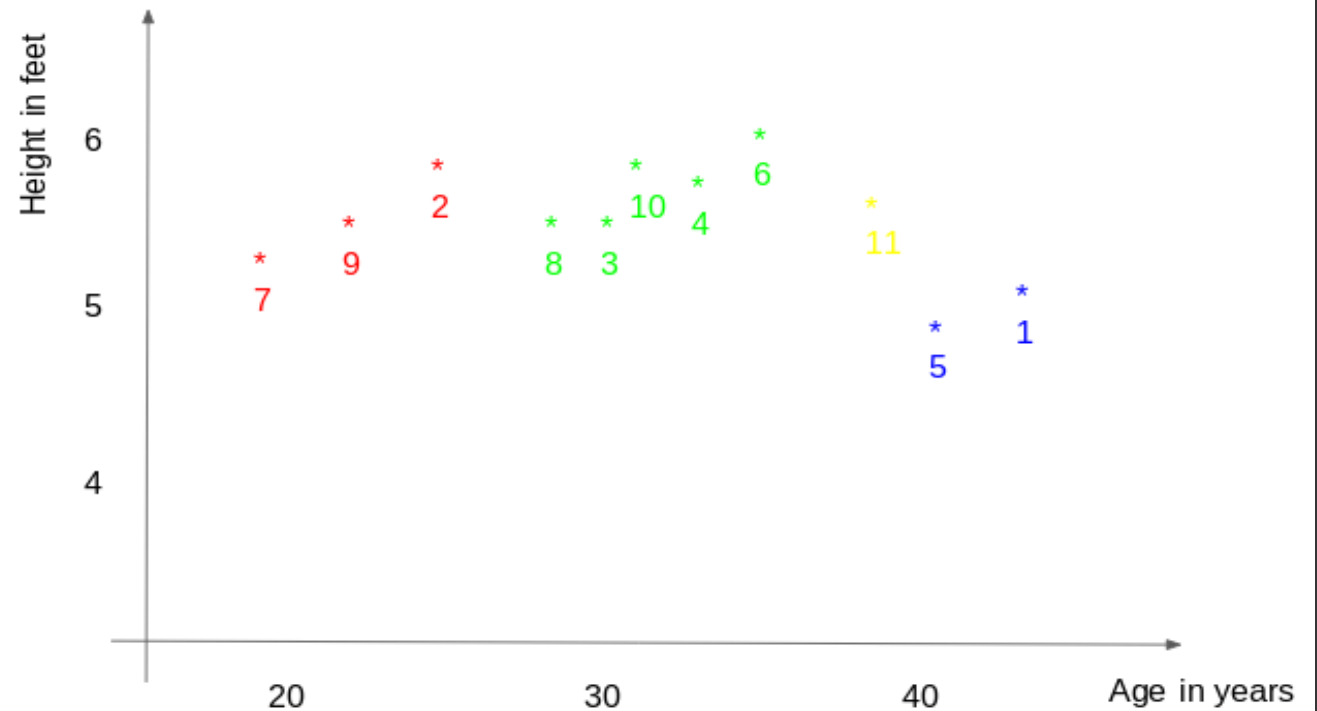
ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	?

Weight:

Less Than 50 is **Low**

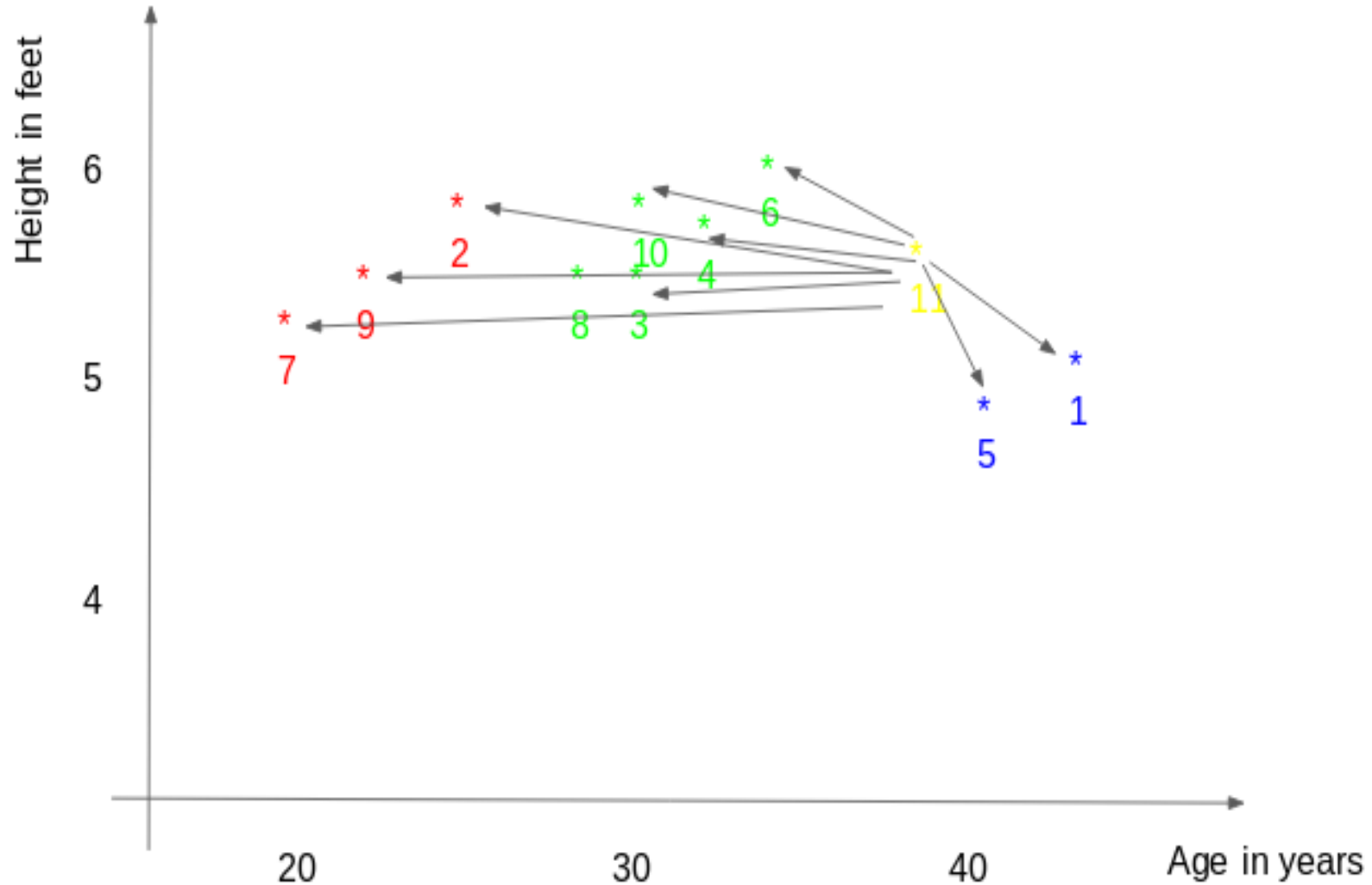
Between 50 to 70 is **MEDIUM**

Greater than 70 is **High**



How kNN works? Step 1

1. Calculate the distance between the new point and each training point is calculated



Methods of Calculating Distance

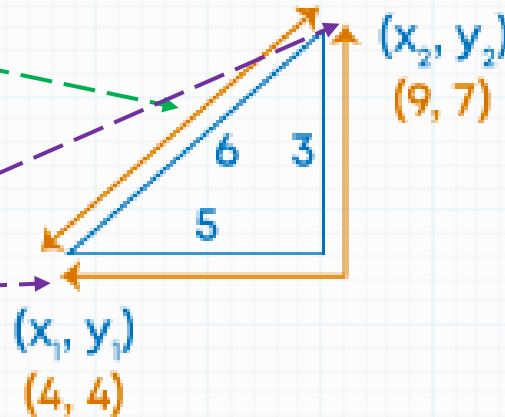
Euclidean

$$\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$

Manhattan

$$\sum_{i=1}^k |x_i - y_i|$$

Example:



Euclidean distance

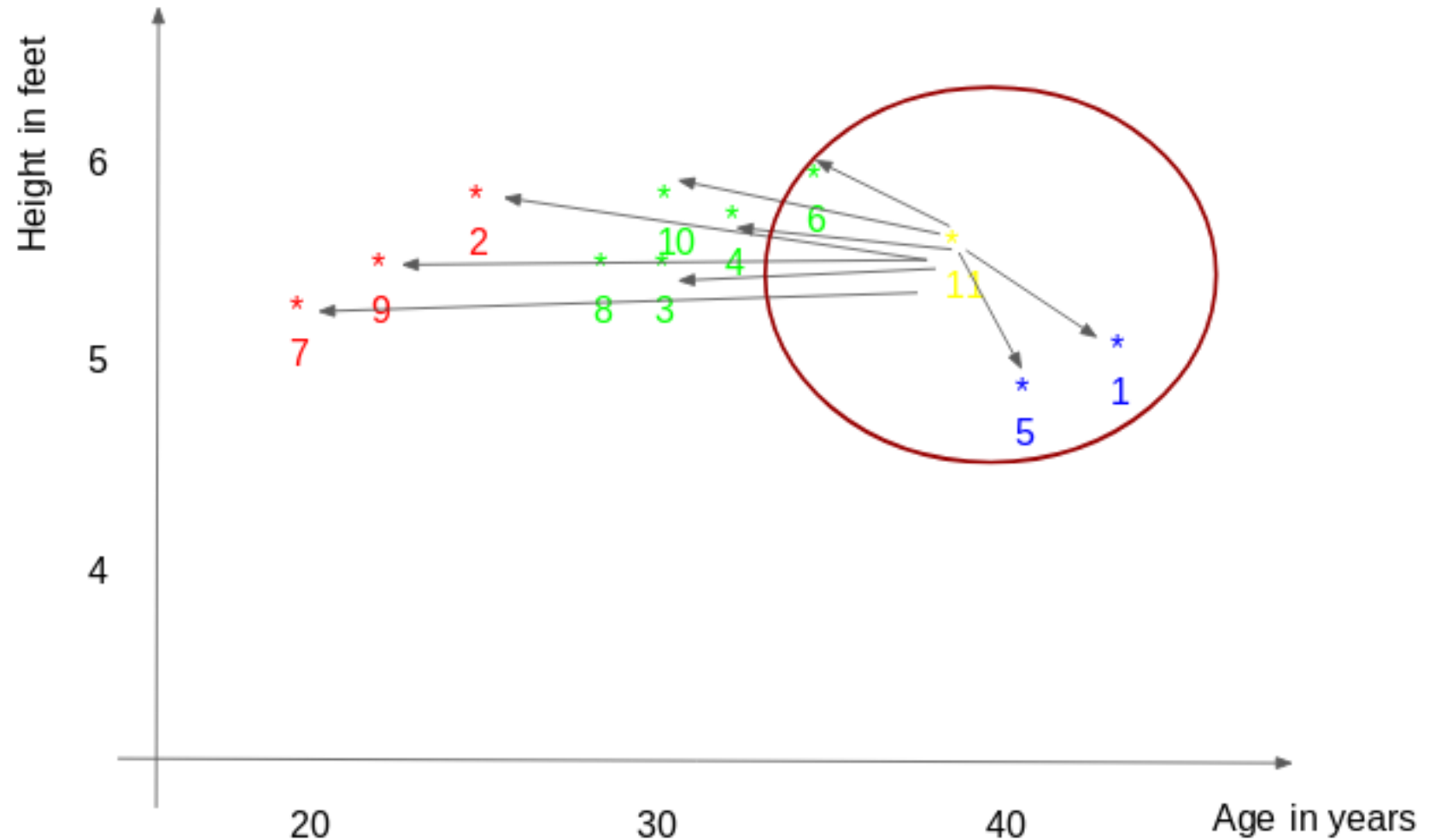
$$\begin{aligned} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(9 - 4)^2 + (7 - 4)^2} \\ &= \sqrt{5^2 + 3^2} \\ &= \sqrt{25 + 9} \\ &= \sqrt{34} \\ &= 5.83 \end{aligned}$$

Manhattan distance

$$\begin{aligned} &= |x_2 - x_1| + |y_2 - y_1| \\ &= |9 - 4| + |7 - 4| \\ &= 5 + 3 \\ &= 8 \end{aligned}$$

How kNN works? Step 2

2. The closest k data points are selected (based on the distance). In this example, points 1, 5, 6 will be selected if the value of k is 3.



How kNN works? Step 3

3. The average of these data points is the final prediction for the new point. Here, we have weight of ID11 = $(77+72+60)/3 = 69.66$ kg.

ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	?

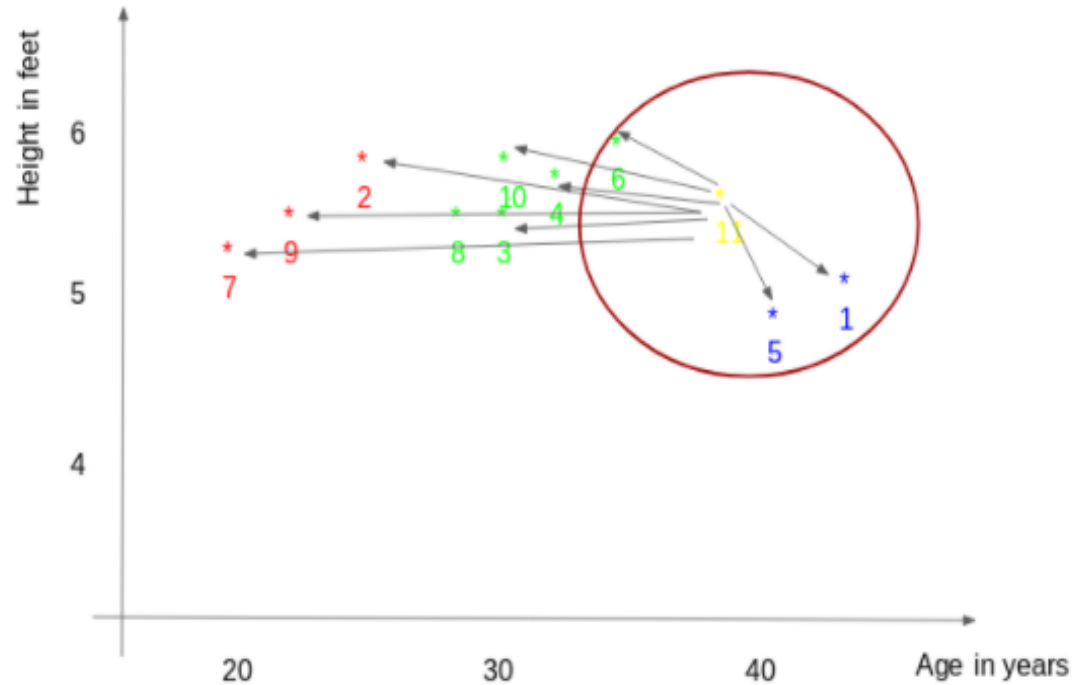
Weight:

Less Than 50 is **Low**

Between 50 to 70 is **Medium**

Greater than 70 is **High**

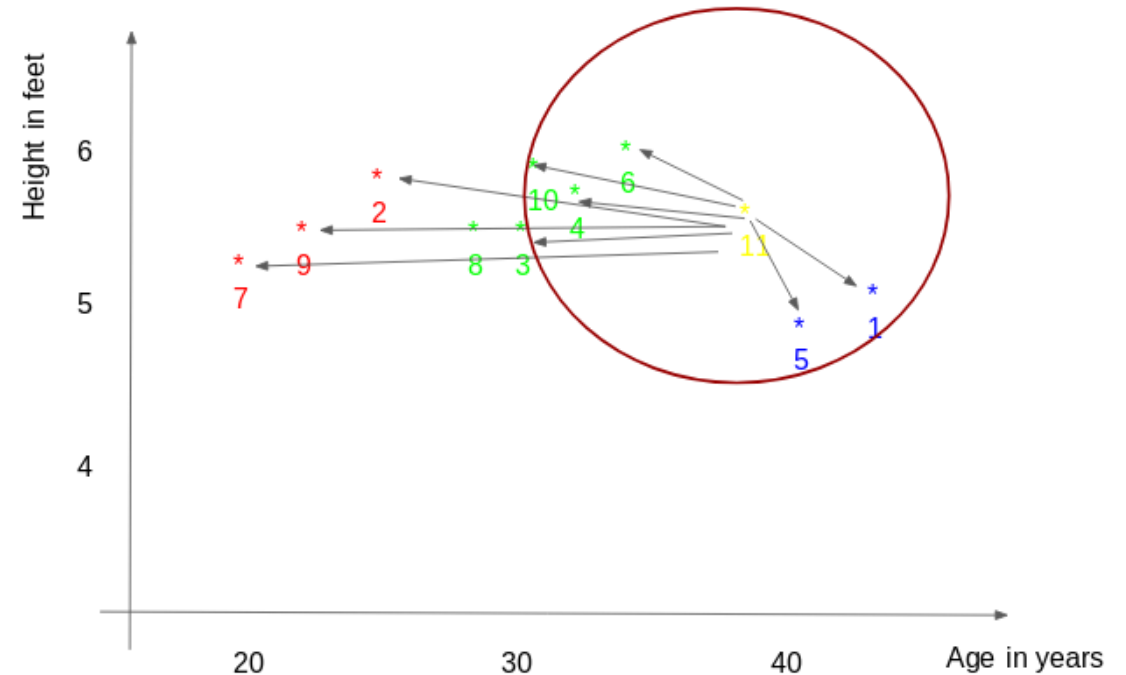
How to choose the factor k?



ID	Height	Age	Weight
1	5	45	77
5	4.8	40	72
6	5.8	36	60

The average of these data points is the final prediction for the new point. Here, we have weight of

$$\text{ID11} = (77+72+60)/3 = 69.66 \text{ kg (Medium Class)}$$

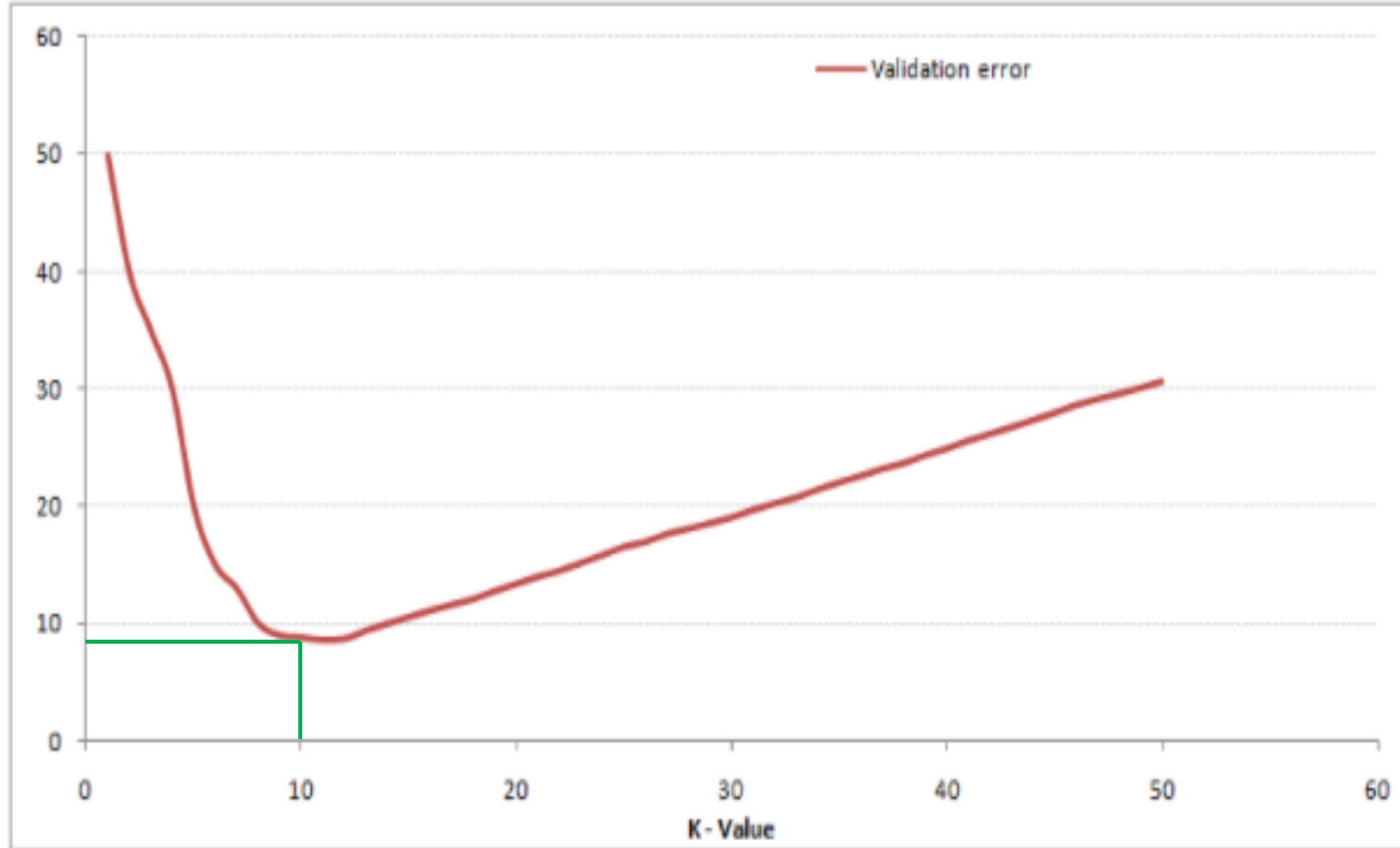


ID	Height	Age	Weight
1	5	45	77
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
10	5.6	32	58

The average of these data points is the final prediction for the new point. Here, we have weight of

$$\text{ID11} = (77+59+72+60+58)/5 = 65.2 \text{ kg (Medium Class)}$$

How to choose the factor k?



- For a very low value of k (suppose $k=1$), the **model overfits** on the **training data**, which leads to a high error rate on the validation set.
- On the other hand, for a high value of k, the **model performs poorly** on both **train and validation set**.

When to Use kNN?



We can use KNN when

Data is labeled



Dog

Dataset is small



Data is noise free

Because KNN is a 'lazy learner' i.e. doesn't learn a discriminative function from the training set

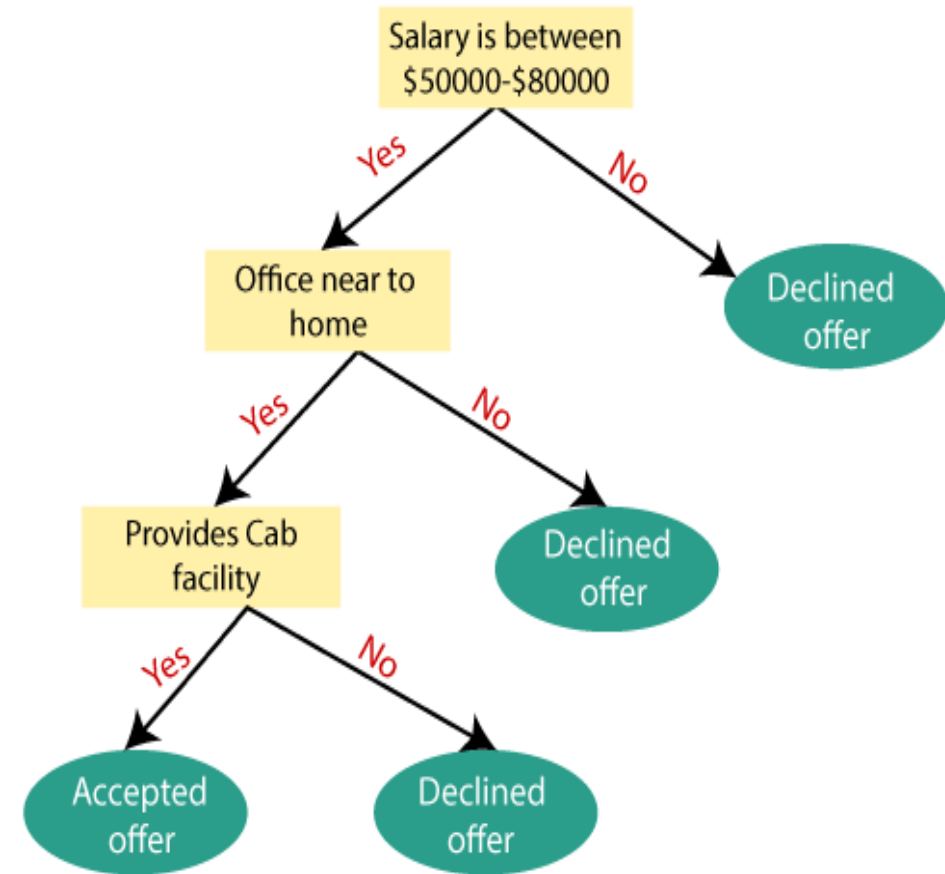
Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	one-fourty
69	176	23
64	173	hello kitty
65	172	Normal

} Noise

BREAK

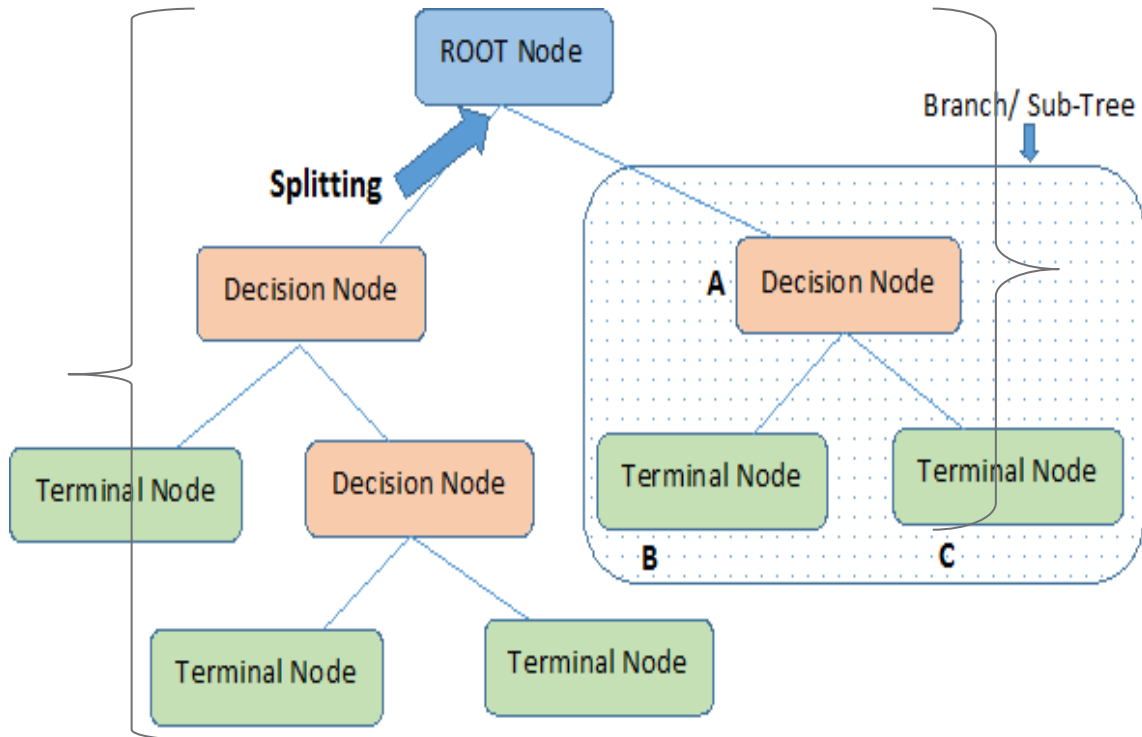
What is a Decision Tree?

- Decision tree is a type of **supervised learning algorithm** (having a predefined target variable) that is mostly used in classification problems.
- Decision Tree is a **white box type** of ML algorithm. It shares internal decision-making logic, which is not available in the **black box type** of algorithms such as Neural Network.
- The decision tree is a **distribution-free or non-parametric method**, which does not depend upon probability distribution assumptions.
- It works for both categorical and continuous input and output variables.
- In this technique, we split the population or sample into two or more homogeneous sets (or sub-populations) based on most significant splitter / differentiator in input variables.



Decision Trees Terminology

Depth =3

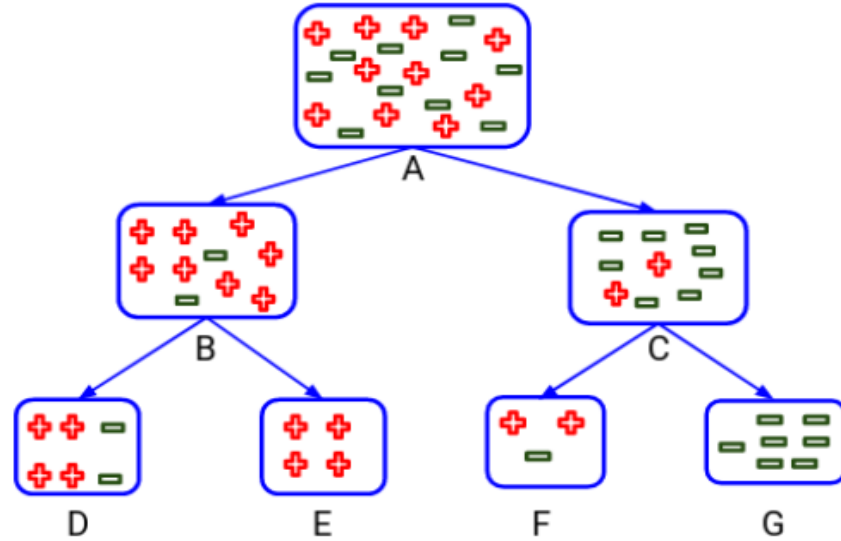


Note:- A is parent node of B and C.

- **Root Node:** It represents entire population or sample and this further gets divided into two or more homogeneous sets.
- **Decision Node:** When a sub-node splits into further sub-nodes, then it is called decision node.
- **Leaf/Terminal Node:** Nodes do not split is called Leaf or Terminal node.
- **Parent and Child Node:** A node, which is divided into sub-nodes is called parent node of sub-nodes where as sub-nodes are the child of parent node.
- **Branch/Sub-Tree:** A sub section of entire tree is called branch or sub-tree.
- **Splitting:** It is a process of dividing a node into two or more sub-nodes.

Quiz

Consider the following tree:



What are the child nodes with respect to Node C?

Choose only ONE best answer.

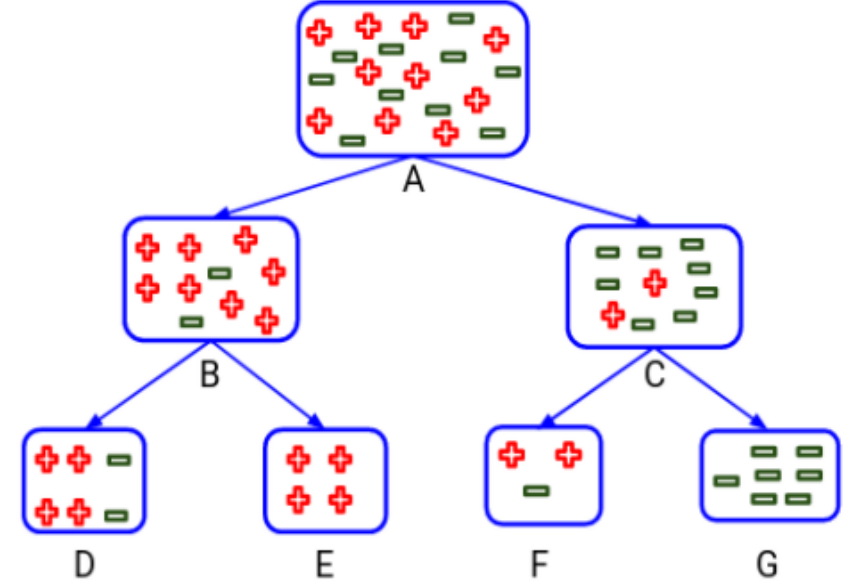
☐ A Only A

☒ Only F and G

☐ C Nodes A, F and G

☐ D None of the options are correct

Consider the following tree:



What is the depth of this tree?

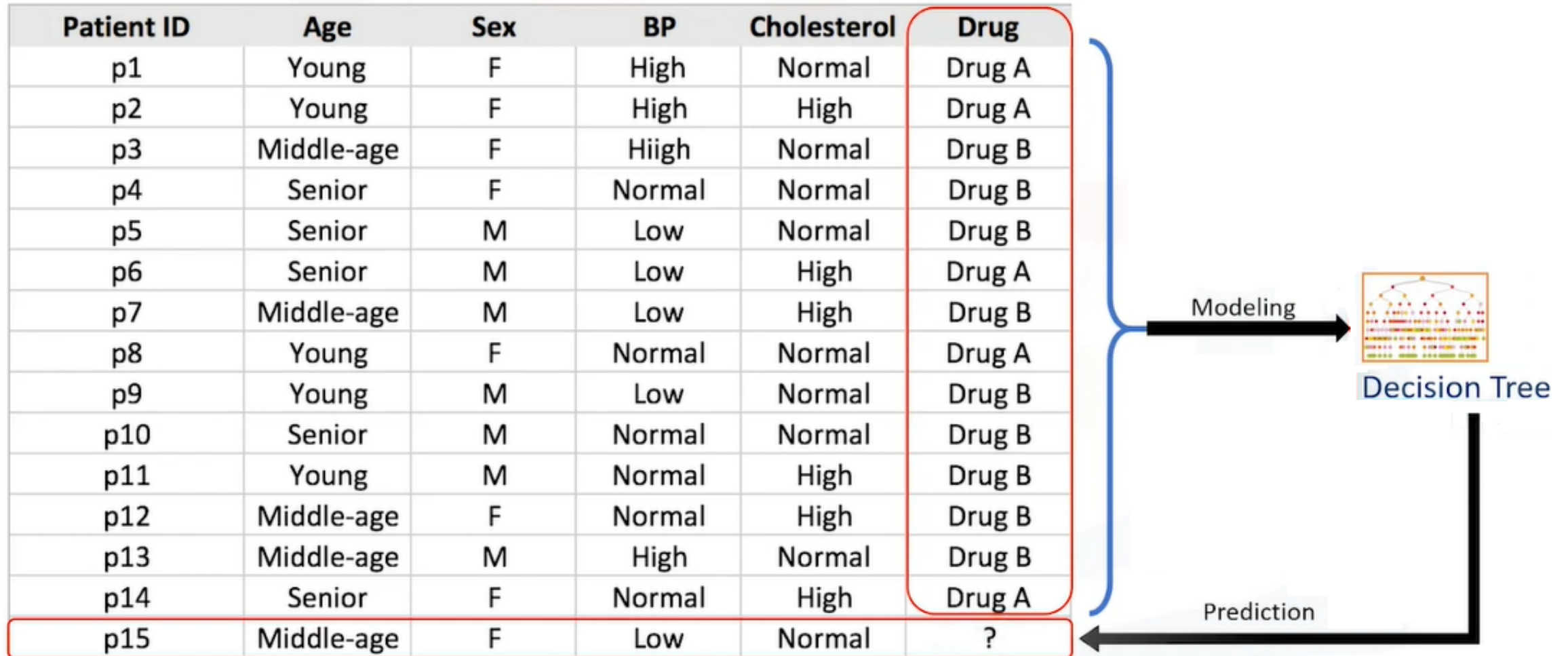
Choose only ONE best answer.

☐ A 4

☐ B 3

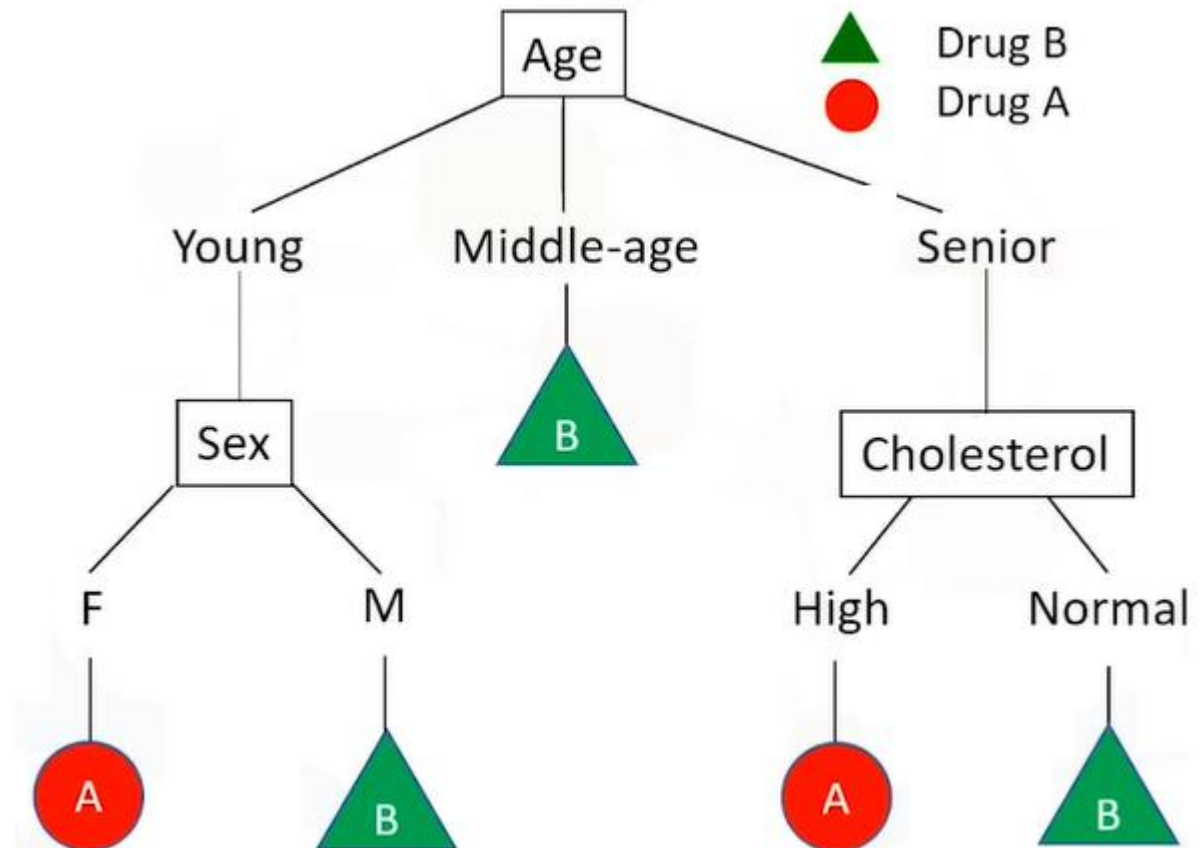
☒ 2

How to Build a Decision Tree?



Building a Decision Tree with Training Set

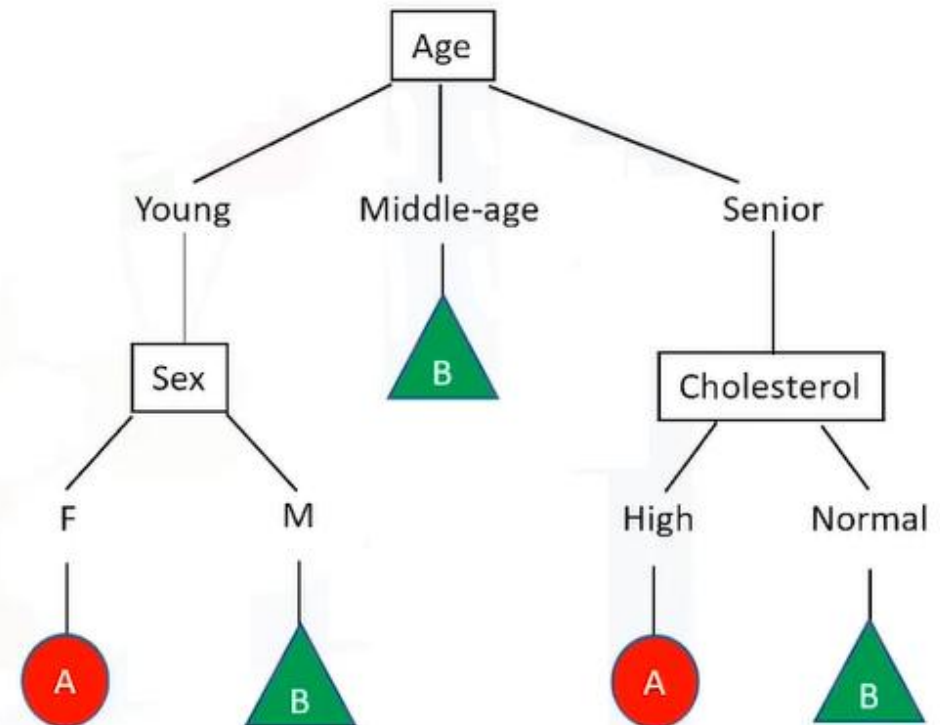
1. Choose an attribute from your dataset.
2. Calculate the significance of attribute in splitting of data.
3. Split data based on the value of the best attribute.
4. Go to step 1.



Building a Decision Tree

Remember: Decision Trees are built using recursive partitioning to classify the data.

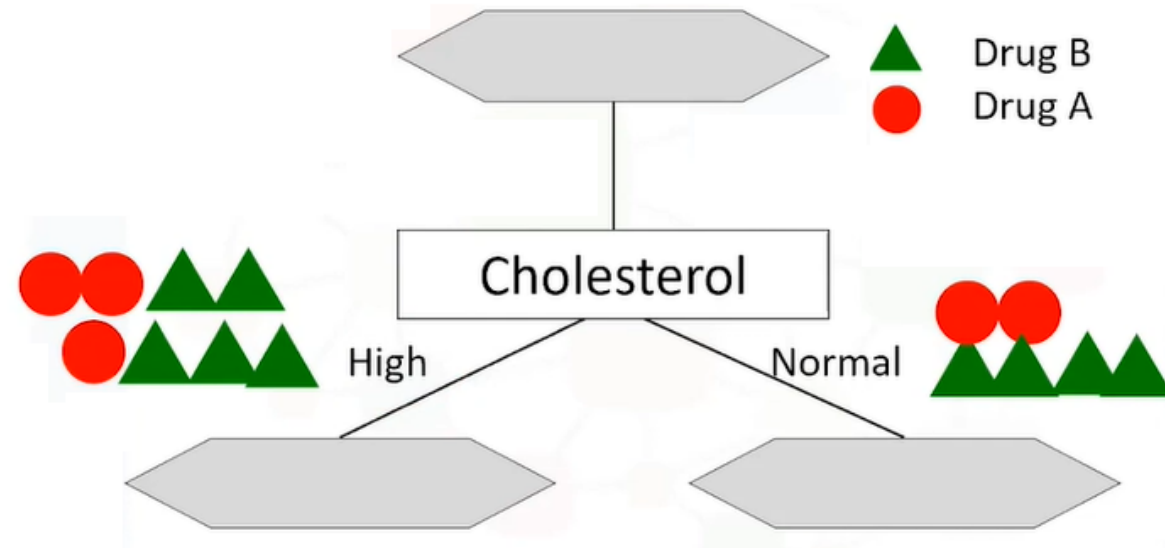
Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	Hiigh	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A
p15	Middle-age	F	Low	Normal	?



Which Attribute is Best?

Remember: Decision Trees are built using recursive partitioning to classify the data.

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	High	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A
p15	Middle-age	F	Low	Normal	?

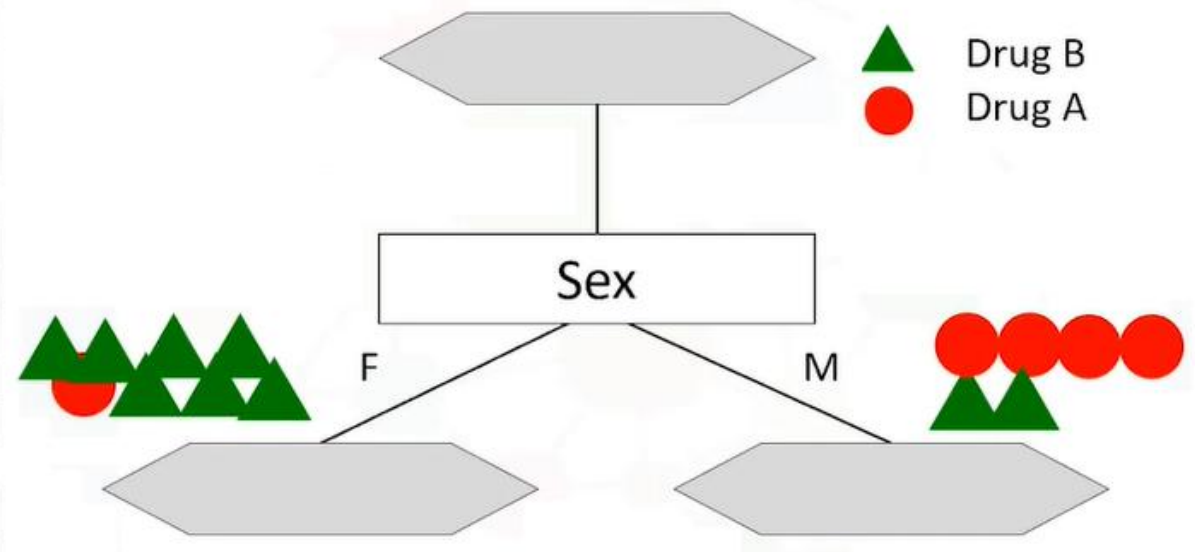


Nodes are not pure, so it is a bad attribute for splitting.

Which Attribute is Best?

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	Hiigh	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A
p15	Middle-age	F	Low	Normal	?

Nodes are more pure, so it is a good attribute for splitting.



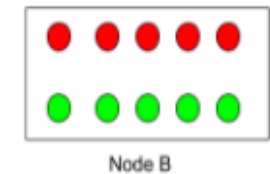
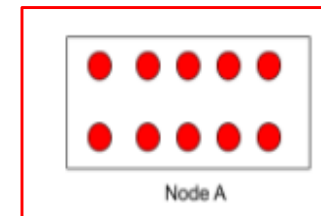
Quiz

What is the basic idea behind choosing the best split in the data?

Choose only ONE best answer.

- A Resultant nodes should be as heterogeneous as possible
-  B Resultant nodes should be as Homogeneous as possible
- C The resultant nodes should have the exact same number of observations

Which of the following is a more pure node?

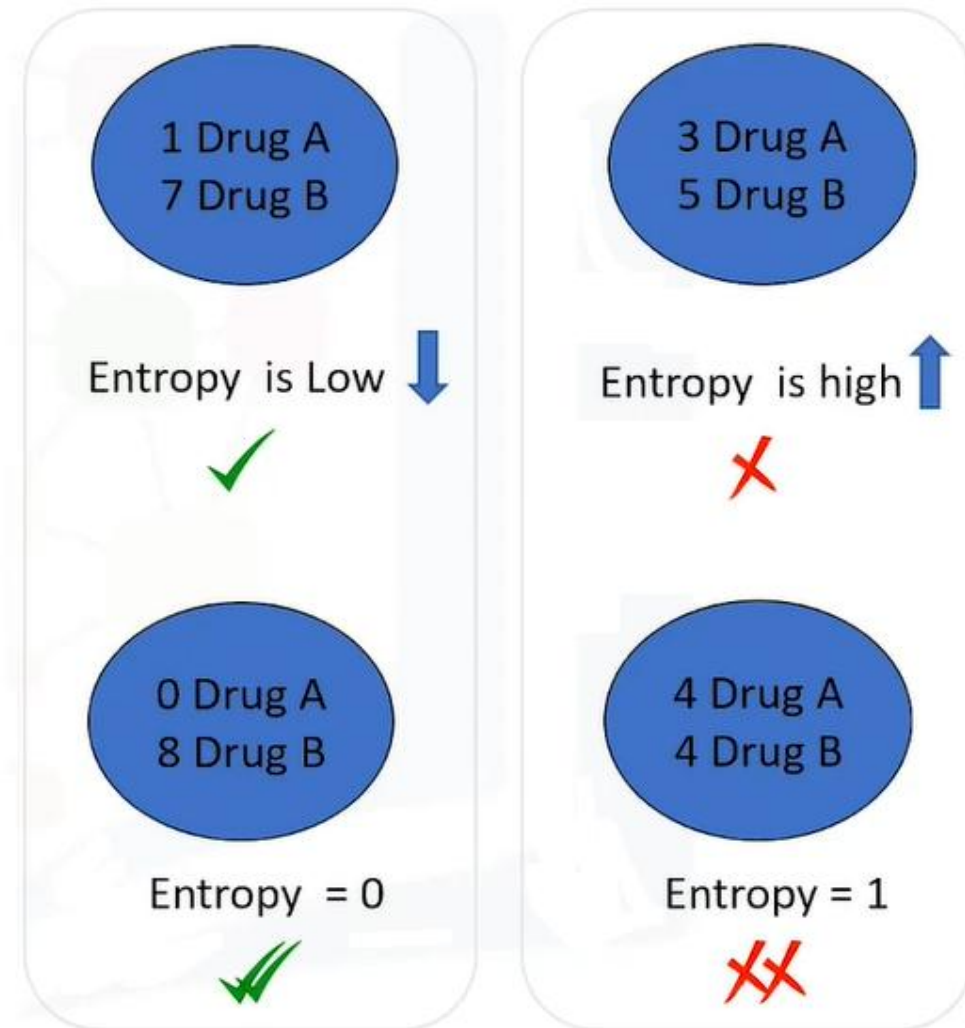


What is Entropy?

- Measure of randomness or uncertainty

$$\text{Entropy} = -p(A)\log(p(A)) - p(B)\log(p(B))$$

The lower the Entropy, the less uniform the distribution, the purer the node.



Remember: You don't have to calculate the entropy as it is easily calculated by packages.

Which Attribute is Best?

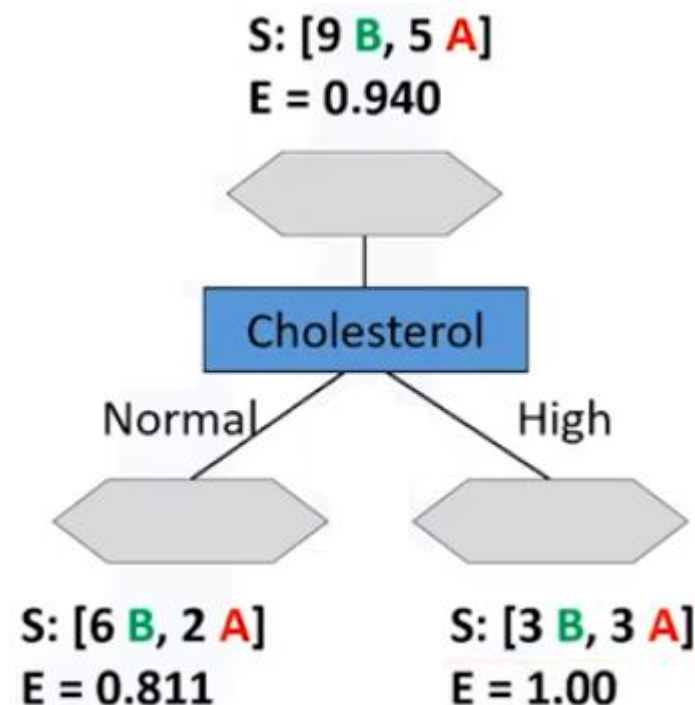
Patient ID	Age	Sex	BP	Cholesterol	Drug
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p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A

S: [9 **B**, 5 **A**] **log base 2**

$$E = -p(B)\log(p(B)) - p(A)\log(p(A))$$

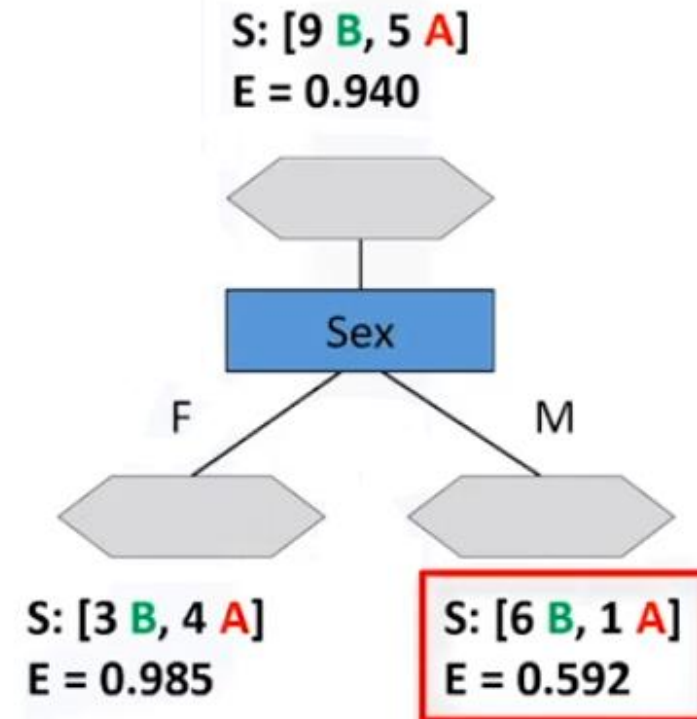
$$E = -(9/14)\log(9/14) - (5/14)\log(5/14)$$

$$E = 0.940$$



Which Attribute is Best?

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	High	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A



Which Attribute is Best?

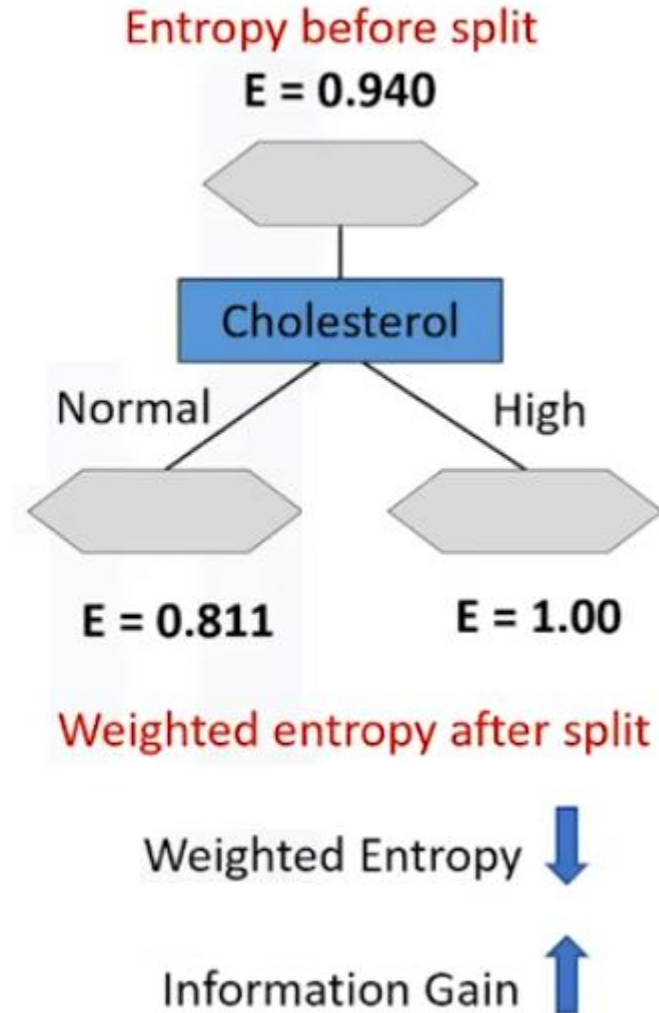


The tree with the higher **Information Gain** after splitting.

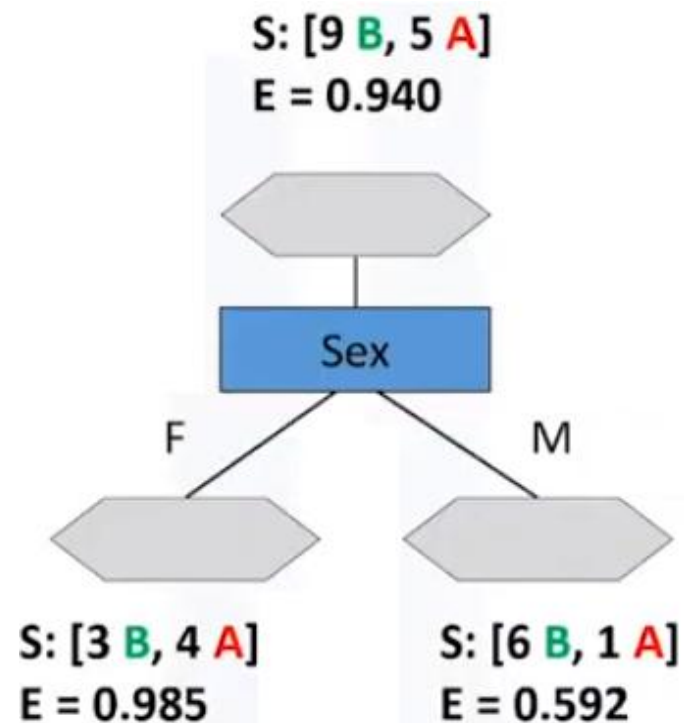
What is Information Gain?

Information gain is the information that can increase the level of certainty after splitting.

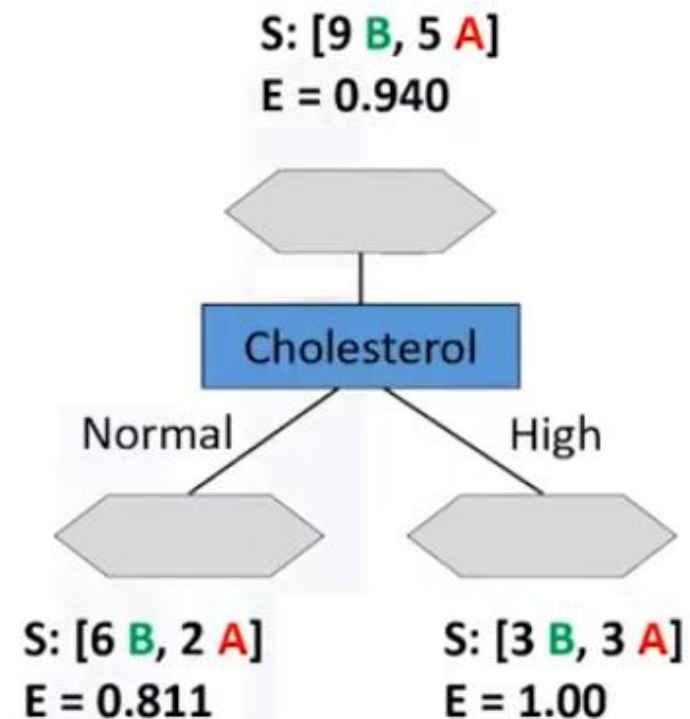
Information Gain = (Entropy before split) – (weighted entropy after split)



Which Attribute is Best?



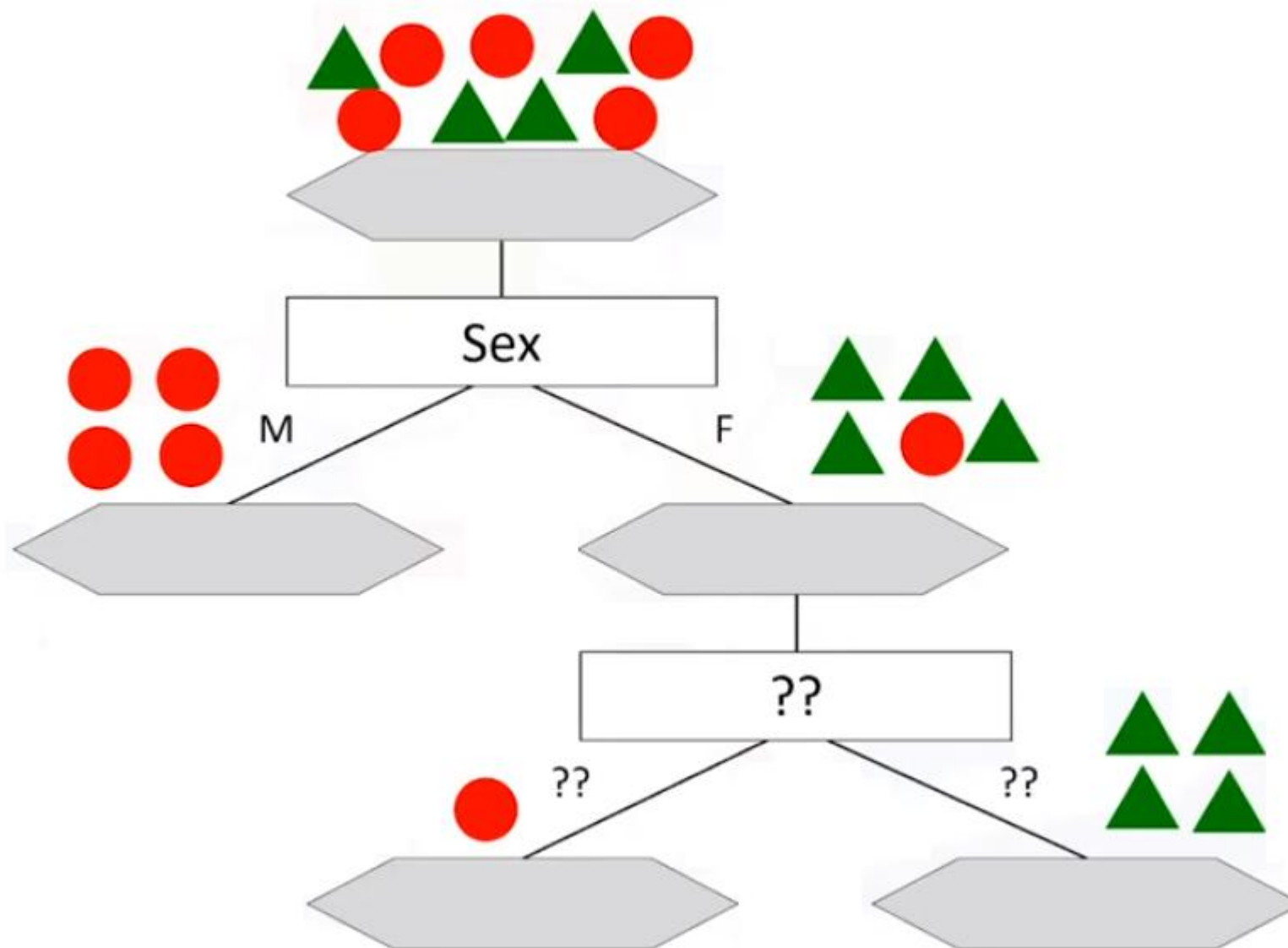
Patient ID	Age	Sex	BP	Cholesterol	Drug
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p3	Middle-age	F	High	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A



$$\begin{aligned} \text{Gain (s, Sex)} &= 0.940 - [(7/14)0.985 + (7/14)0.592] \\ &= 0.151 \end{aligned}$$

$$\begin{aligned} \text{Gain (s, Cholesterol)} &= 0.940 - [(8/14)0.811 + (6/14)1.0] \\ &= 0.048 \end{aligned}$$

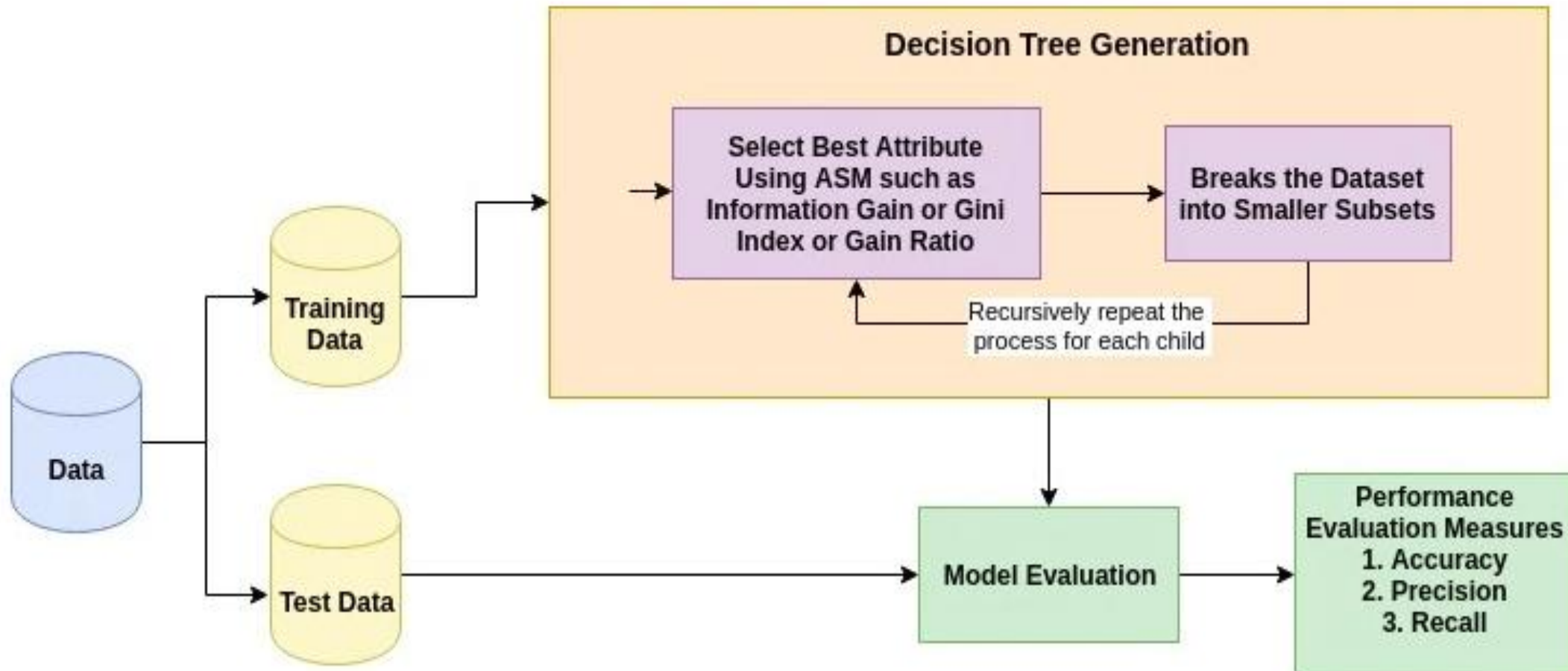
Building a Decision Tree



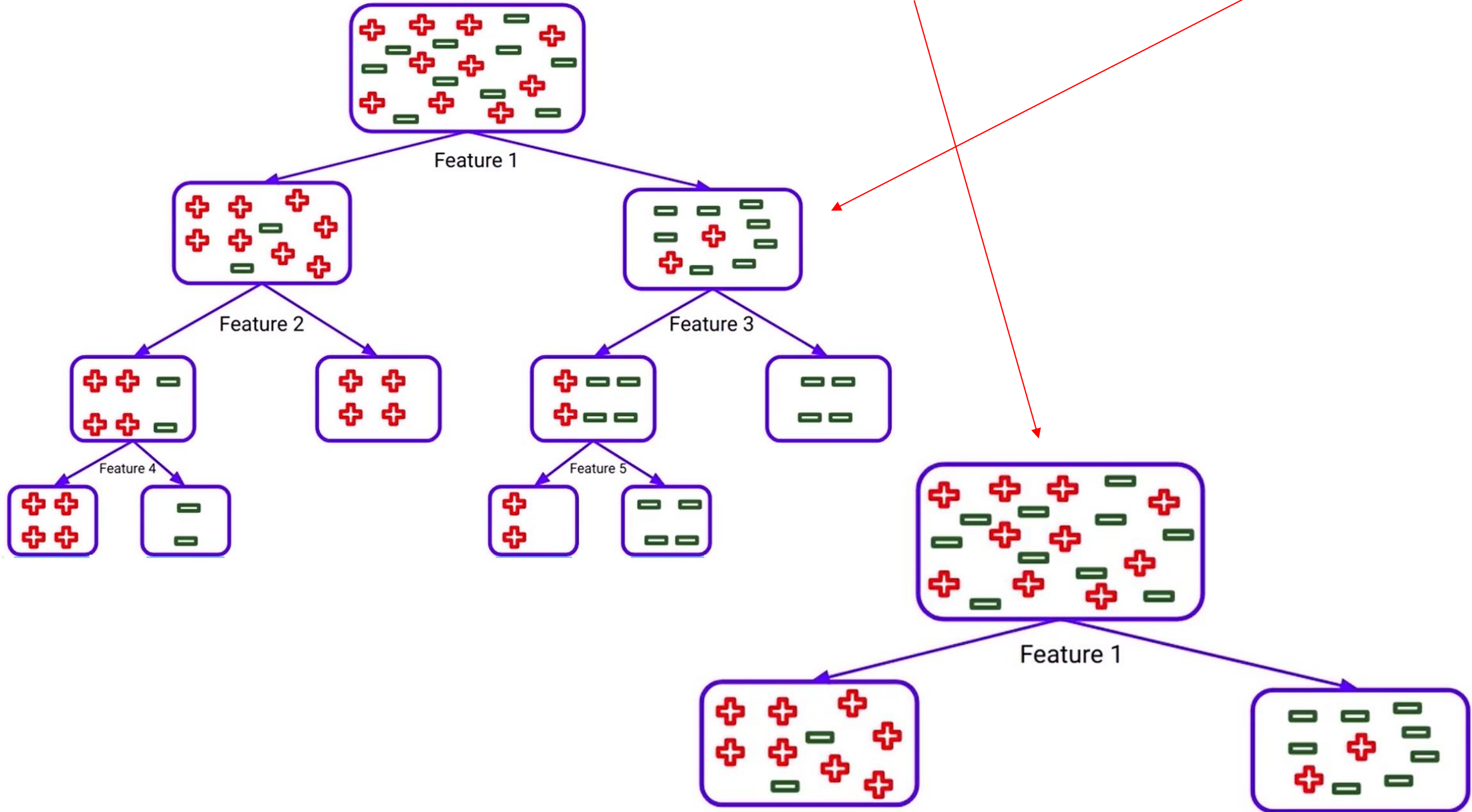
BREAK

How Does the Decision Tree Algorithm Work?

Attribute Selection Measures (ASM) to split the records.



Underfitting and Overfitting



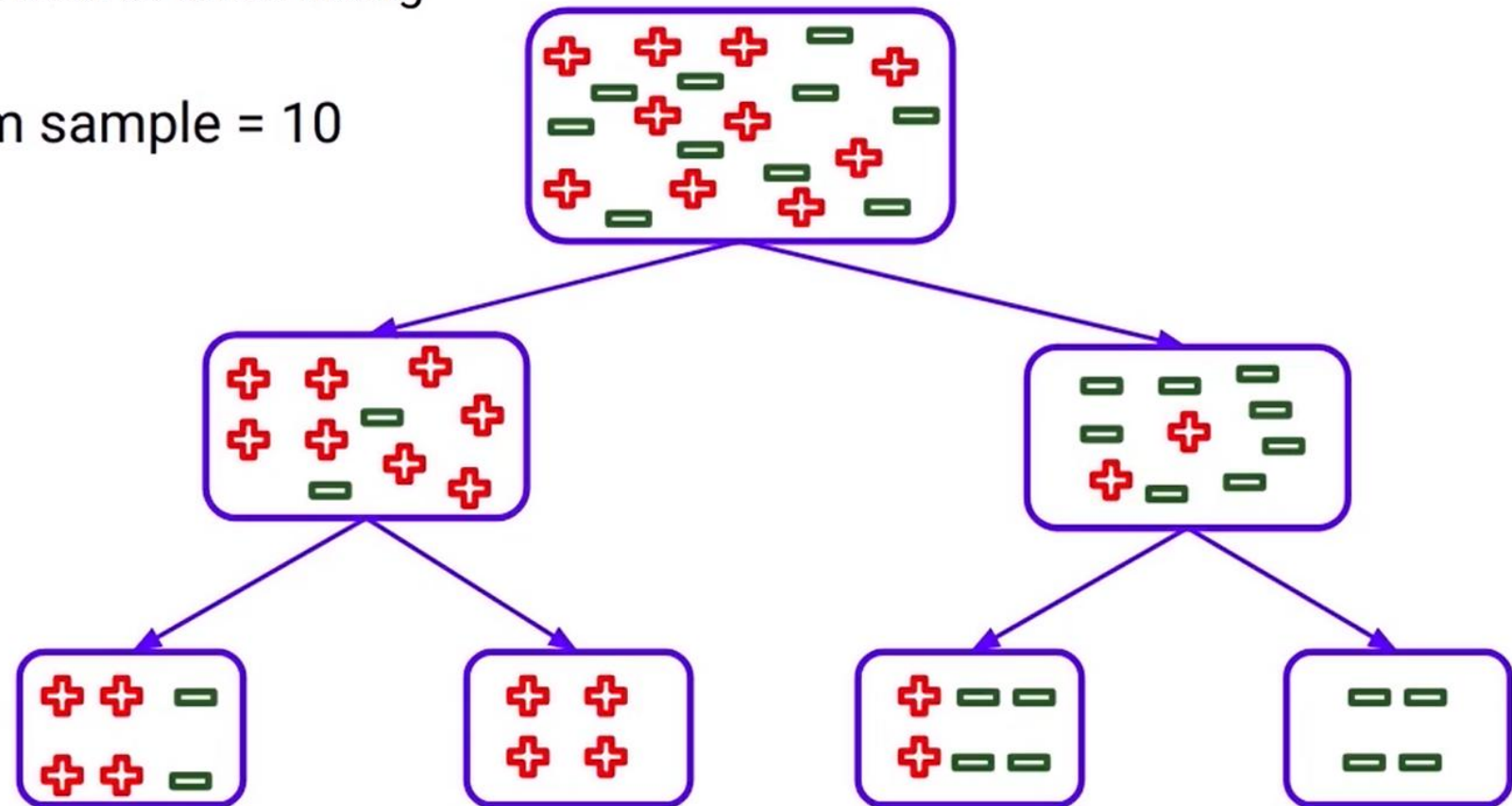
Optimizing Performance of Decision Trees

1. Minimum samples for a node split
2. Minimum samples for a terminal node
3. Maximum depth of tree
4. Maximum number of terminal nodes

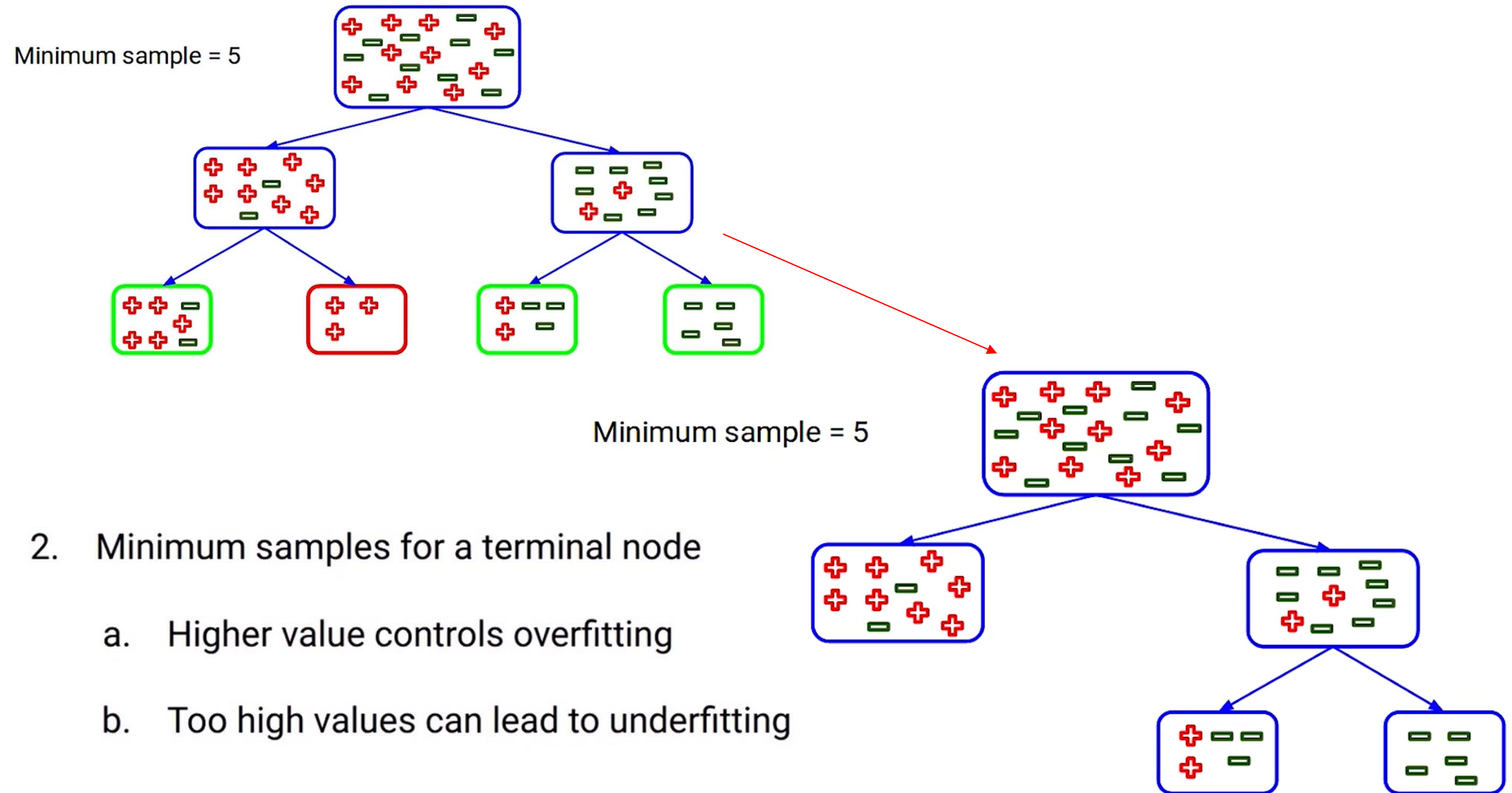
Strategies to Avoid Underfitting and Overfitting

1. Minimum samples for a node split
 - a. Higher values controls overfitting
 - b. Too high values can lead to underfitting

Minimum sample = 10

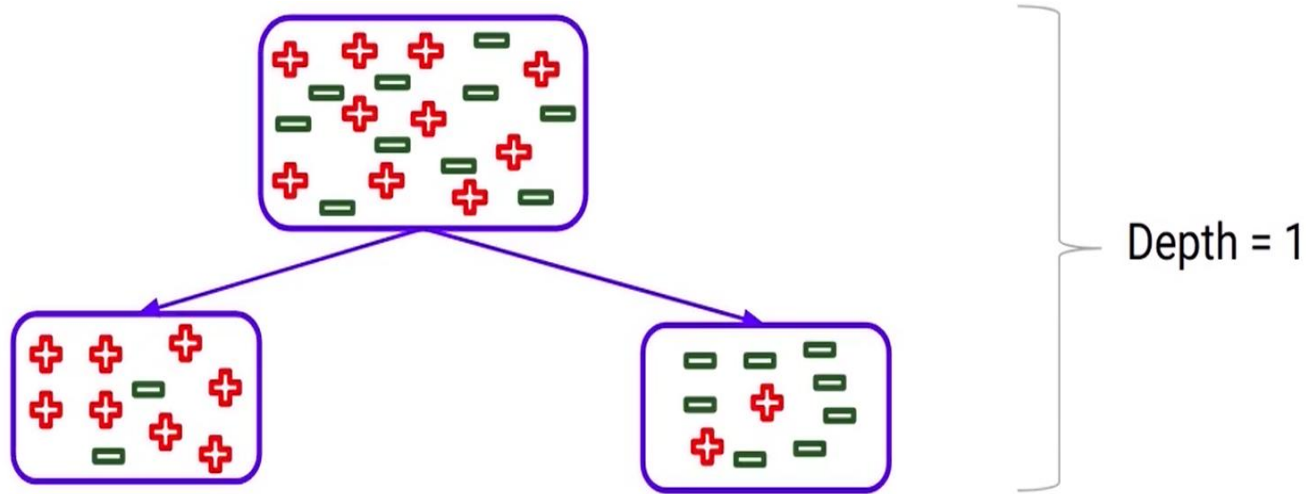


Strategies to Avoid Underfitting and Overfitting

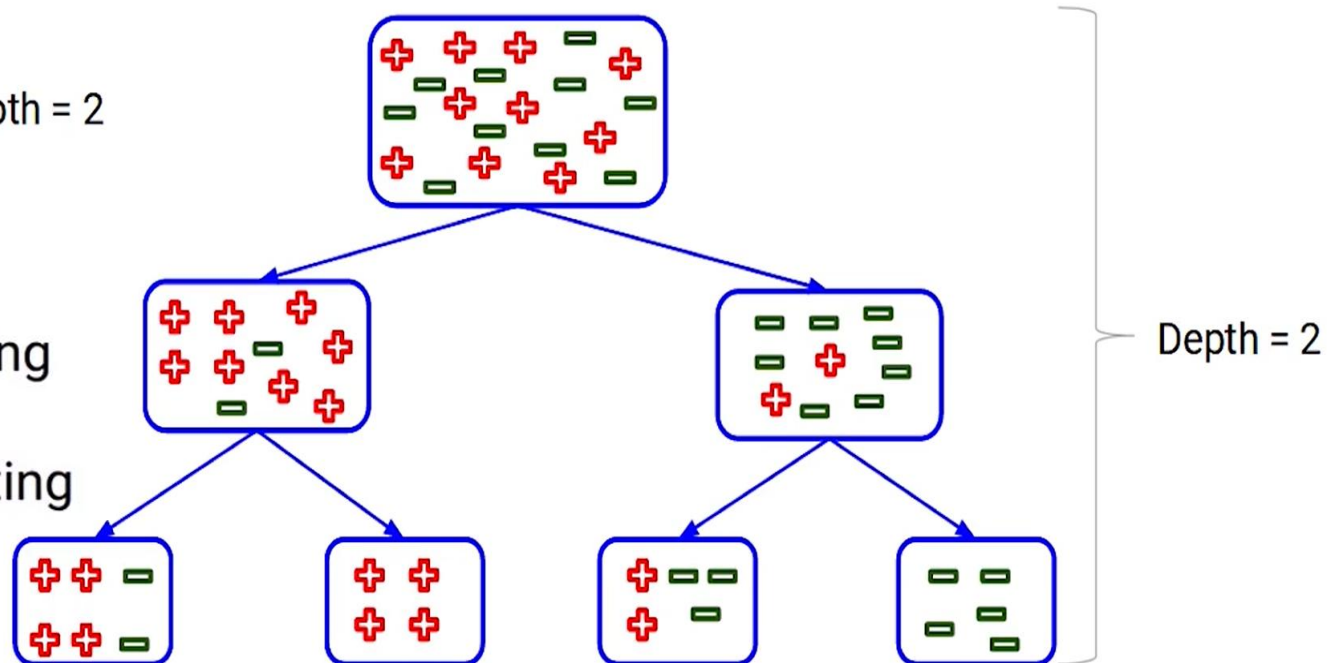


Strategies to Avoid Underfitting and Overfitting

Max Depth = 2



Max Depth = 2

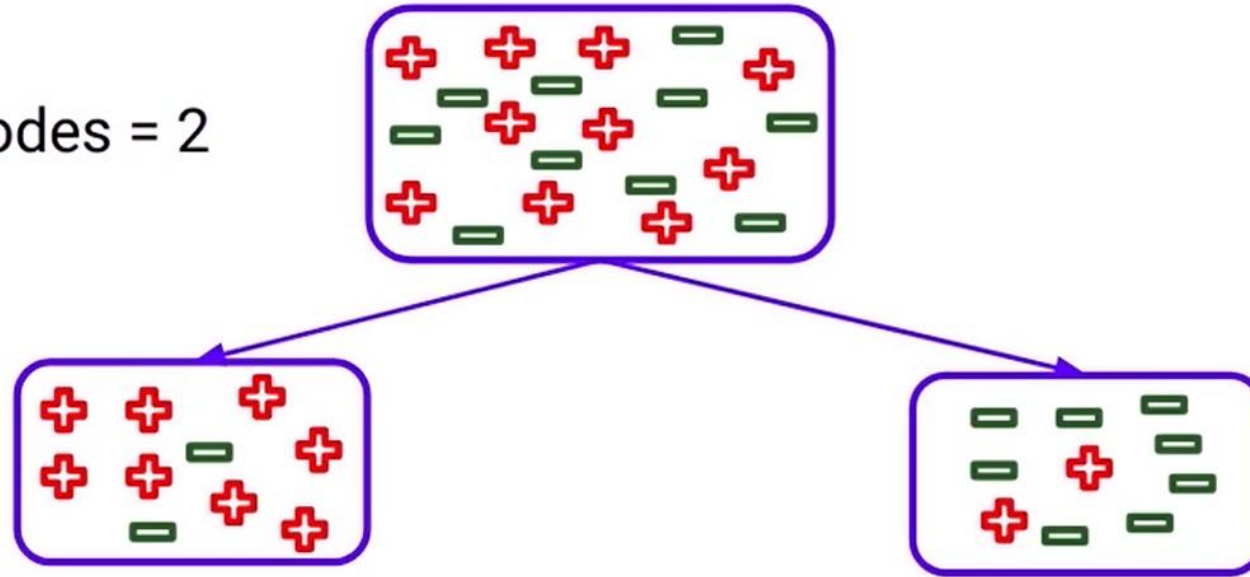


3. Maximum depth of tree

- a. Higher depth can lead to overfitting
- b. Lower depth can lead to underfitting

Strategies to Avoid Underfitting and Overfitting

Max Terminal nodes = 2



Quiz

Which of the following parameter(s) can help to prevent overfitting?

Choose ALL answers that apply.



Max depth



Minimum samples for node split



Minimum samples for leaf node

D

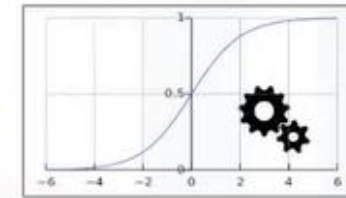
Splitting criterion

BREAK

Model Evaluation Metrics

	tenure	age	address	income	ed	employ	equip	callcard	wireless	churn
3	11.0	33.0	7.0	136.0	5.0	5.0	0.0	1.0	1.0	1
1	33.0	33.0	12.0	33.0	2.0	0.0	0.0	0.0	0.0	1
2	23.0	30.0	9.0	30.0	1.0	2.0	0.0	0.0	0.0	0
3	38.0	35.0	5.0	76.0	2.0	10.0	1.0	1.0	1.0	0
4	7.0	35.0	14.0	80.0	2.0	15.0	0.0	1.0	0.0	0
5	68.0	52.0	17.0	130.0	1.0	24.0	0.0	0.0	0.0	0
6	42.0	40.0	7.0	37.0	3.0	8.0	1.0	0.0	0.0	0
7	8.0	21.0	1.2	17.0	3.0	2.0	0.0	0.0	0.0	0
8	36.0	50.0	26.0	140.0	2.0	21.0	0.0	0.0	0.0	0
9	49.0	51.0	27.0	63.0	4.0	19.0	0.0	0.0	0.0	0

Train set



Predicted churn

1

1

0

1

0

Actual Labels y

$\hat{y}$ Predicted Labels

Test

Test set

	tenure	age	address	income	ed	employ	equip	callcard	wireless	churn
0	11.0	33.0	7.0	136.0	5.0	5.0	0.0	1.0	1.0	1
1	33.0	33.0	12.0	33.0	2.0	0.0	0.0	0.0	0.0	1
2	23.0	30.0	9.0	30.0	1.0	2.0	0.0	0.0	0.0	0
3	38.0	35.0	5.0	76.0	2.0	10.0	1.0	1.0	1.0	0
4	7.0	35.0	14.0	80.0	2.0	15.0	0.0	1.0	0.0	0

Jaccard Index

y : Actual labels

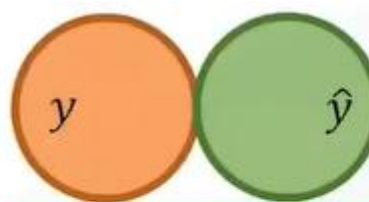
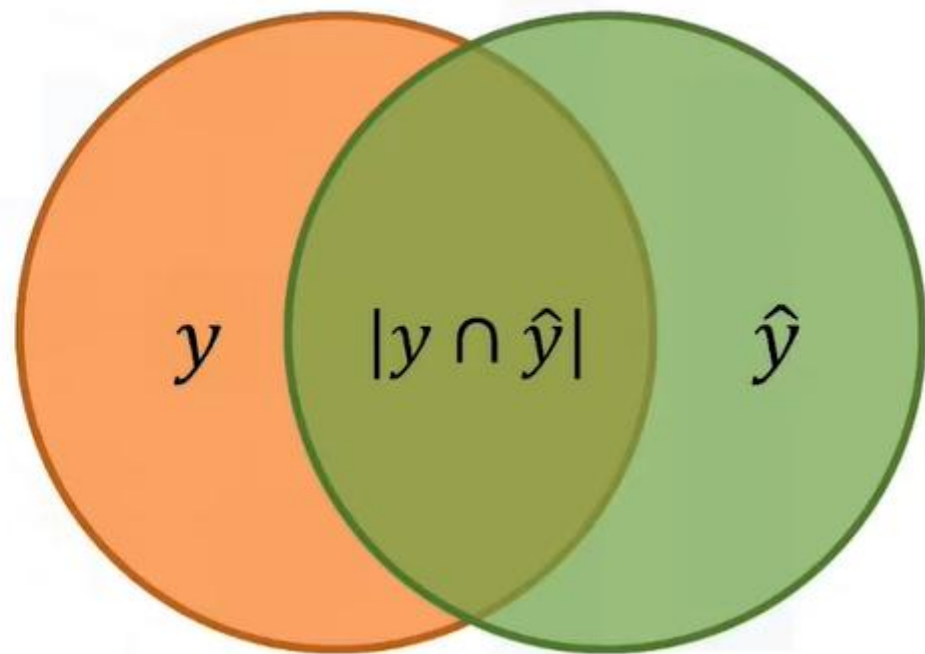
$\hat{y}$: Predicted labels

$$J(y, \hat{y}) = \frac{|y \cap \hat{y}|}{|y \cup \hat{y}|} = \frac{|y \cap \hat{y}|}{|y| + |\hat{y}| - |y \cap \hat{y}|}$$

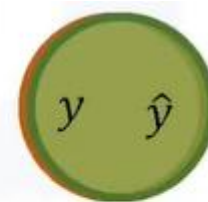
y : [0, 0, 0, 0, 0, 1, 1, 1, 1, 1]

$\hat{y}$: [1, 1, 0, 0, 0, 1, 1, 1, 1, 1]

$$J(y, \hat{y}) = \frac{8}{10+10-8} = 0.66$$



$$J(y, \hat{y}) = 0.0$$



$$J(y, \hat{y}) = 1.0$$

Higher Accuracy

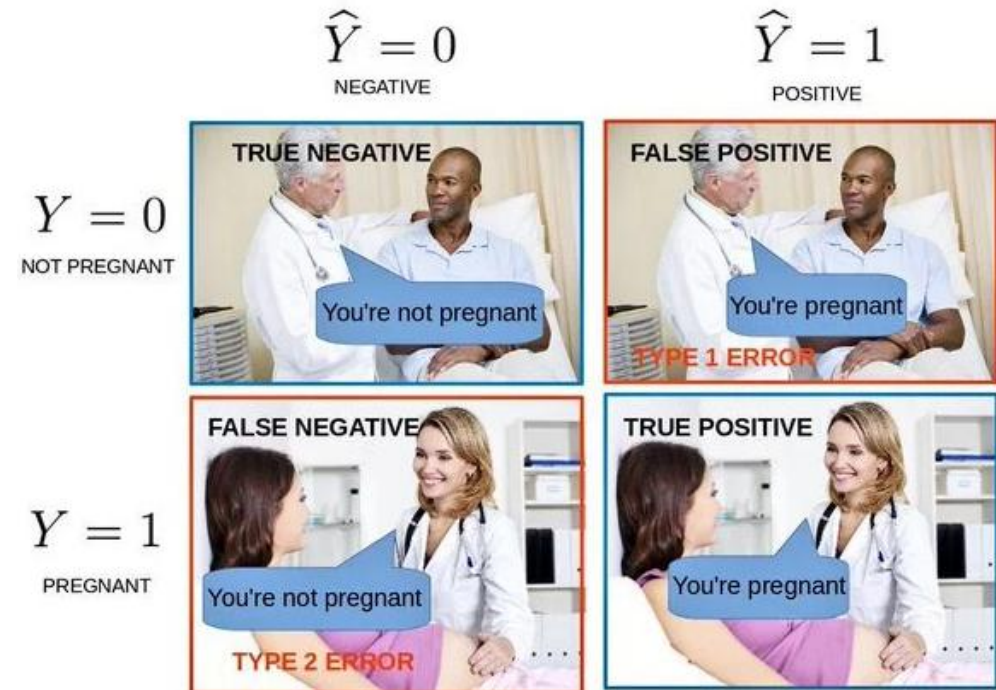


Confusion Matrix

Confusion Matrix

		Actual Value	
		Positive	Negative
Predicted Value	Positive	TP (True Positive)	FP (False Positive)
	Negative	FN (False Negative)	TN (True Negative)

- True Positive (TP) : Observation is positive, and is predicted to be positive.
- False Negative (FN) : Observation is positive, but is predicted negative.
- True Negative (TN) : Observation is negative, and is predicted to be negative.
- False Positive (FP) : Observation is negative, but is predicted positive.



Model Evaluation Metrics

The equations of 4 key classification metrics

Accuracy:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

F₁ score:

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

<code>precision_recall_curve(y_true, probas_pred, *)</code>	Compute precision-recall pairs for different probability thresholds
<code>roc_curve(y_true, y_score, *, pos_label, ...)</code>	Compute Receiver operating characteristic (ROC)

Others also work in the multiclass case:

<code>balanced_accuracy_score(y_true, y_pred, *, ...)</code>	Compute the balanced accuracy
<code>cohen_kappa_score(y1, y2, *, labels, ...)</code>	Cohen's kappa: a statistic that measures inter-annotator agreement.
<code>confusion_matrix(y_true, y_pred, *, ...)</code>	Compute confusion matrix to evaluate the accuracy of a classification.
<code>hinge_loss(y_true, pred_decision, *, ...)</code>	Average hinge loss (non-regularized)
<code>matthews_corrcoef(y_true, y_pred, *, ...)</code>	Compute the Matthews correlation coefficient (MCC)
<code>roc_auc_score(y_true, y_score, *, average, ...)</code>	Compute Area Under the Receiver Operating Characteristic Curve (ROC AUC) from prediction scores.

Some also work in the multilabel case:

<code>accuracy_score(y_true, y_pred, *, ...)</code>	Accuracy classification score.
<code>classification_report(y_true, y_pred, *, ...)</code>	Build a text report showing the main classification metrics.
<code>f1_score(y_true, y_pred, *, labels, ...)</code>	Compute the F1 score, also known as balanced F-score or F-measure
<code>fbeta_score(y_true, y_pred, *, beta[, ...])</code>	Compute the F-beta score
<code>hamming_loss(y_true, y_pred, *, sample_weight)</code>	Compute the average Hamming loss.
<code>jaccard_score(y_true, y_pred, *, labels, ...)</code>	Jaccard similarity coefficient score
<code>log_loss(y_true, y_pred, *, eps, ...)</code>	Log loss, aka logistic loss or cross-entropy loss.
<code>multilabel_confusion_matrix(y_true, y_pred, *)</code>	Compute a confusion matrix for each class or sample
<code>precision_recall_fscore_support(y_true, ...)</code>	Compute precision, recall, F-measure and support for each class
<code>precision_score(y_true, y_pred, *, labels, ...)</code>	Compute the precision
<code>recall_score(y_true, y_pred, *, labels, ...)</code>	Compute the recall
<code>roc_auc_score(y_true, y_score, *, average, ...)</code>	Compute Area Under the Receiver Operating Characteristic Curve (ROC AUC) from prediction scores.
<code>zero_one_loss(y_true, y_pred, *, ...)</code>	Zero-one classification loss.

And some work with binary and multilabel (but not multiclass) problems:

<code>average_precision_score(y_true, y_score, *)</code>	Compute average precision (AP) from prediction scores
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Mentimeter Quiz



Mentimeter

Get Ready to Test Your Knowledge

Thank You
for
Attention

